The Case of Apple's Restrictions on Super Apps: Is the U.S. Department of Justice Right?

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October 13, 2026

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Abstract

The novel "super app" technology, which emerged in the early 2010s, enables app developers to run a single version of an app on different operating systems. Apple's alleged practice of restricting super apps in the United States is one of the subjects of a recent antitrust complaint filed by the U.S. Department of Justice (2024a). I present a model of a two-sided market with cross-side network effects to investigate two questions: why Apple restricts super apps in the United States but not in Asia; and whether its restrictions may adversely affect market competition and welfare in the United States. I find that a primary reason for Apple's behavior is the dominant position of its iOS platform on the consumer side of the U.S. smartphone platform market—iOS does not hold this position in Asia. Furthermore, I show that whether these restrictions harm consumers and developers depends critically on whether consumers view platforms as highly differentiated or close substitutes, the extent to which restrictions on super apps increase consumers' switching costs and decrease developers' multi-homing savings, and the number of apps operating on each platform.

1 Introduction

"Apple has disrupted the growth of apps with broad functionality that would make it easier for consumers to switch between competing smartphone platforms" (U.S. Department of Justice, 2024b). In March 2024, an antitrust complaint filed by the U.S. Department of Justice (2024a) argued that Apple has been behaving in an anti-competitive manner by

*I am especially grateful to Thomas Wiseman, David Sibley, V. Bhaskar, and John Hatfield for helpful comments, suggestions, and encouragement. Thanks are also due to Svetlana Boyarchenko, Vasiliki Skreta, Maxwell Stinchcombe, Caroline Thomas, Rafael Veiel, and the participants in The University of Texas at Austin theory writing seminar for their helpful feedback. All remaining errors and omissions are my own.

"suppressing super apps." Having emerged in the early 2010s, super apps differ from other types of apps in that they do not merely serve consumers. They also operate as infrastructure for third-party app developers (referred to as developers thereafter) by hosting other third-party apps. In particular, "developers can write mini programs that run on the super app without having to write separate apps for iPhones and other smartphones" (U.S. Department of Justice, 2024a). According to the U.S. Department of Justice (2024a), "Since at least 2017, Apple has arbitrarily imposed exclusionary requirements that unnecessarily and unjustifiably restrict mini programs and super apps."

For instance, Apple has "required apps in the United States to display mini programs using a flat, text-only list of mini programs," "banned displaying mini programs with icons or tiles," "banned apps from categorizing mini programs," and "selectively enforced its contractual rules with developers to prevent developers from monetizing mini programs" (U.S. Department of Justice, 2024a). These restrictions imposed by Apple constitute refusals to deal with mini apps—and therefore with super apps: on its iOS platform, Apple restricts mini apps hosted by super apps from accessing the full set of functionalities—including custom UI display and in-app monetization—granted to standalone applications.

In contrast, Apple seldom exhibits these behaviors in Asia. This contrast corresponds to the regional difference in the position of iOS on the consumer side of the smartphone platform market—that is, the market share of iOS, or equivalently, the fraction of consumers who use iOS in the smartphone platform market: iOS holds a dominant position in the United States but not in Asia.[1]

Correspondingly, my analysis sheds analytical light on the following two questions: why Apple has been restricting super apps in the United States but not in Asia; and whether Apple's restrictions may adversely affect market competition and welfare in the United States. In particular, I find that a key reason for Apple to restrict super apps in the U.S. is the dominant position of its iOS, and such restrictions harm its rival platforms. Meanwhile, the welfare effects of such restrictions depend on whether consumers view platforms as highly differentiated or close substitutes, the extent to which restrictions on super apps increase consumers' switching costs and decrease developers' multi-homing savings, and the number of apps operating on each platform.

To address these two research questions, I model the smartphone platform market by

[1]In terms of mobile operating system market share, as of June 2026, iOS accounted for approximately 30.79% of the global market; 58.35% of the U.S. market; and 21.69% of the Asian market (in particular, 29.13% in China, 6.86% in India, 22.32% in Indonesia, and 45.68% in Singapore). In contrast, as of the same period, Android accounted for approximately 69.14% of the global market; 41.62% of the U.S. market; and 78.22% of the Asian market (in particular, 70.52% in China, 93.05% in India, 77.6% in Indonesia, and 54.28% in Singapore). Data source: Statcounter Global Stats (2026), which treats tablet operating systems and mobile operating systems as parallel segments instead of nesting the former under the latter.

embedding super apps and platforms' initial market shares into the the seminal two-sided-market model of Armstrong (2006). I model operating systems—Apple's iOS and Google's Android—as platforms competing for two distinct groups of user: consumers on one side and developers on the other.[2] In such a two-sided market, I focus on the developer-supporting functionality of super apps rather than treating them as platforms that compete with operating systems for users.

Instead, super apps affect the smartphone platform market via the following two channels. First, they reduce consumers' switching costs. With super apps, when switching to a new platform, a consumer finds it easier—that is, less costly—to adapt to using the apps on that platform. This is due to the same versions of mini apps available on super apps that operate on both his or her original platform and the new platform.[3] Second, they reduce the costs borne by developers who multi-home, because they allow developers operating on both platforms to reuse the same app resources across platforms. Each of the two channels through which super apps operate affects one side of the market directly—consumer switching costs on one side, developer multi-homing cost reductions on the other—and the opposite side indirectly through cross-side network effects.

The market power that enables Apple to restrict super apps in the smartphone platform market comes from its full control over the iOS App Store. The App Store is the unique channel for developers to get access to consumers on the iOS platform and is fully controlled by Apple in any region.[4] Previous studies have found that Apple has extensively used this market power to restrict third-party apps and favor its own apps, even though such conduct harms welfare on both sides of the market (Geradin and Katsifis, 2021; Wörsdörfer, 2024). However, in the case of super apps, Apple has only used this market power in the United States but not in Asia. This regional variation in Apple's attitude toward super apps corresponds to that in iOS's position on the consumer side of the smartphone platform market. In other words, Apple turns out to have been restricting super apps in regions where iOS is strong—that is, where iOS holds a dominant position—on the consumer side, but not in those where iOS is weak.

I capture platforms' positions on the consumer side by assuming that, at the beginning of

[2]In my analysis, I focus on consumers' payoffs from using apps and abstract from any utility associated with phone services.

[3]For mini-apps, their functionalities are aligned with those of ordinary apps published directly on operating systems, while they are "developed using programming languages such as HTML5 and JavaScript", which are "standard in most web pages" (U.S. Department of Justice, 2024a). Consequently, this feature enables developers to write a single version of a mini app that can, via the channel of super apps, serve any consumer regardless of the operating system his or her smartphone uses.

[4]In contrast, Android is open-source, and developers can bypass the Google Play Store and distribute applications independently.

the game, each consumer is a patron of one of the two platforms. This assumption constitutes the platforms' exogenous initial market shares. The platform whose initial market share is larger than its rival is strong—equivalently, stronger than its rival—on the consumer side of the market.

I do not treat super apps as players in the game. Instead, I incorporate them exogenously into the model via the two channels described above: super apps reduce both consumers' switching costs and the costs of developers who multi-home. This is because, despite Apple's restrictions, super apps remain available across both platforms, enhancing cross-platform interoperability and reducing ecosystem frictions for both consumers and developers. Correspondingly, the effects of restrictions on super apps also operate through the above two channels, but in directions opposite to those of super apps.

The model also incorporates cross-side network effects: consumers are better off when more developers operate on the platforms they respectively use, and developers are better off when more consumers use the platforms they respectively operate on.[5] Such effects capture the fundamental trade-offs platforms face when setting prices across market sides: each platform has an incentive to set a relatively low price on one side to expand participation there and increase the surplus available for extraction on the other side, where it can charge a relatively high price for the sake of extraction.

To understand the effects of restrictions on super apps, I first examine the impact of the following two factors that result from such restrictions: the reduction in multi-homers' cost savings and the increase in consumers' switching costs. I find that when the cost savings of multi-homing developers increase, consumers and platforms are better off, while developers are worse off. This is because both platforms proportionally raise their developer-side prices in order to extract the additional developer surplus that arises from multi-homers' higher cost savings. I show that when consumers' switching costs increase, a platform becomes better off if and only if it is strong on the consumer side. The effect on consumers depends on how horizontally differentiated they are. When their preferences for their preferred platform are sufficiently strong, consumer surplus decreases in the switching costs. In contrast, when consumers are not very horizontally differentiated, an increase in consumers' switching costs may make consumers better off.

The above findings, plus the fact that restrictions on super apps raise consumers' switch-

[5]In contrast, in the model, I assume away same-side network effects. Thus, regardless of which platform a user chooses, the user's payoff is invariant to the number of same-side users on that platform. In the presence of consumers' switching costs, same-side network effects that are symmetric across platforms simply reinforce the dominant position of the platform that is already strong on the consumer side. Hence, abstracting from same-side network effects allows the analysis to isolate how cross-side network effects interact with the other forces in the model, disentangling these mechanisms from those associated with same-side network effects.

ing costs and reduce multi-homers' cost savings, yield the following results. A platform that is weak on the consumer side, such as iOS—and thus its owner, Apple—in Asia, does not prefer such restrictions. However, the preference of a strong platform, such as iOS in the United States, depends on the extent to which restrictions on super apps increase consumers' switching costs and decrease developers' multi-homing savings—an increase in the former raises the profit of such a strong platform, while a reduction in the latter lowers it. The overall effect on consumers may be positive if (1) restrictions on super apps lead to a substantial increase in consumers' switching costs against a comparatively minor decrease in developers' multi-homing savings and (2) consumers view the platforms as sufficiently close substitutes. Otherwise, consumers are worse off.

These results point to a central reason for the regional variation in Apple's attitude toward super apps: iOS is strong on the consumer side of the smartphone platform market in the United States but not in Asia. Restrictions on super apps may be profitable for a platform that is strong on the consumer side, like iOS—and thus, Apple—in the United States, because the restrictions raise consumers' switching costs.

In addition, under an extension of the baseline model in which platforms' developer-side prices are rigid—as is the case empirically[6]—the effect of restrictions on super apps on developer surplus exhibits a pattern similar to that on consumer surplus. Developers may benefit when (1) restrictions on super apps lead to a substantial increase in consumers' switching costs against a comparatively minor decrease in developers' multi-homing savings and (2) consumers view the platforms as sufficiently close substitutes, if in addition (3) the strong platform serves more developers than the weak platform. Otherwise, developers are worse off under the restrictions.

Thus, my analysis identifies three factors that determine the directions of the welfare effects of restrictions on super apps: (1) whether consumers view platforms as highly differentiated or close substitutes, (2) the extent to which restrictions on super apps increase consumer adaptation costs upon switching platforms and heighten developer workloads associated with platform-specific maintenance, and (3) the number of apps operate on each platform. Evaluating potential antitrust concerns associated with Apple's restrictions on super apps in the United States requires closer empirical examination of these relevant factors.

[6]In reality, both Apple and Google (the owner of Android) have long maintained a standard 30% commission alongside fixed lump-sum fees—a one-time $25 payment for Android and annual fees of $99 or $299 for iOS. Although both of them have recently lowered their commission rates, these reductions resulted from antitrust litigation rather than from market forces.

Contribution and Related Literature

My results have direct implications for the practical assessment of antitrust cases. They pertain to both the economic mechanism through which two-sided markets operate and to legal practice. With respect to the economic mechanism, the implications primarily arise from my findings about Apple's use of its full control over the iOS App Store to reinforce its market power over consumers. In previously investigated antitrust cases of Apple (Kotapati et al., 2020; Geradin and Katsifis, 2021; Shao, 2021; Bremond, 2024; Wörsdörfer, 2024), researchers found that Apple usually uses its full control over the App Store to favor its own apps and extract surplus from other developers by charging those developers discriminatory fees and restricting the functionalities of third-party apps. Thus, researchers tend to consider this market power to be mainly on the developer side.

However, in the case of restrictions on super apps, Apple's aim is not to privilege its own competing apps—Apple does not operate a super app—but to prevent any super app from thriving on iOS. Instead, in my model, the effects of this market power fall primarily on the consumer side. In particular, this developer-side market power of Apple originates in its unique control over the access to iOS consumers, and in the case of super apps, this power propagates back to the consumer side through the restrictions Apple imposes. Accordingly, I present a framework for analyzing such situations where a factor may directly affect both sides of a two-sided market. The results of my analysis can also facilitate understanding both Apple's 2024 antitrust case and other analogous cases.[7]

Another aspect of the implications pertains to legal practice. In a CRS (Congressional Research Service) Legal Sidebar, Sykes (2024) suggests that it may be preferable for the U.S. Department of Justice to characterize its Apple lawsuit as a technological-tying case rather than a refusal-to-deal one, given the difficulty plaintiffs face in prevailing on refusal-to-deal claims. Technological tying and refusal to deal refer to distinct forms of restrictions on products: the former refers to situations in which a firm "designs one product so that it functions only when used in conjunction with its own complementary products" (Gaynor, 2006), whereas the latter refers to cases where a firm refuses to do some business with another party in order to maintain or exploit its market power.

In my model, Apple's restrictions on super apps constitute a form of technological tying imposed on both consumers and developers through refusals to deal with super apps. Such restrictions apply uniformly to all super apps and therefore require (1) consumers to use iOS devices to access the versions of apps available on iOS, and (2) developers to use iOS's

[7] E.g., the U.S. versus Microsoft 1998 antitrust case, where Java and non-Microsoft browsers played a role similar to that of super apps in the Apple 2024 antitrust case. For further details, refer to paragraphs 7 and 8 in the complaint filed by the U.S. Department of Justice (1998).

APIs (app programming interfaces) and publish their apps on the iOS App Store in order to better serve iOS consumers. As a result, my model interprets these restrictions as a hybrid of technological tying and refusals to deal.

My results also contribute to the thin strand of literature that incorporates switching costs (Klemperer, 1987a,b, 1988; Farrell and Shapiro, 1988; Taylor, 2003; Toolsema, 2009; Biglaiser et al., 2013, 2016; Fabra and García, 2015) into models of two-sided markets (Armstrong, 2006; Armstrong and Wright, 2007; Hagiu, 2009; Shao, 2016; Carroni et al., 2024).[8] One type of setting commonly used to investigate the effects of switching costs in two-sided markets is explicitly intertemporal. It assumes that platforms enter the two-sided market simultaneously and compete for users in the first period in order to increase their respective market power coming from the switching costs in the second period. Lam (2017) developed a model with exogenous switching costs on both sides of the market. Kim (2017) considered a situation where consumers' switching costs depend on compatibility between the competing platforms. Cavazos Flores (2023) incorporated buyers' imperfect information about platforms' qualities into the setting. A limitation of this type of setting is that platforms' market power in the second period simply depends on variables explicitly incorporated into the model, which cannot account for other realistic factors that lie outside the scope of such variables.

Another type of setting incorporates one of these realistic factors: sequential entry. In such type of setting, one platform arrives earlier than its (prospective) rival(s) and can attract consumers to gain its market power as an incumbent before the arrival of the later entrant(s). Tremblay (2019) adopted a dynamic setting, assuming consumers' switching costs in the second period to be endogenously dependent on the network effect enjoyed in the first period. Giovannetti and Siciliani (2023) opted for a static setting, explicitly assumed that the incumbent platform initially hosts all users in the market, and allowed for differentiated switching costs for both sides of the market. However, in this type of setting, the effect of the incumbent's dominance over the entrant(s) on the market outcomes tends to be relatively overwhelming compared with those of other factors. Thus, this type of setting seems to lack the ability to flexibly capture the sources of platforms' market power that arise from switching costs.

The novel focus of my model is how differences in platforms' initial market shares affect their attitudes toward super apps and the relevant restrictions. Consequently, I explicitly specify the exogenous initial market shares of the platforms and allow for a rich set of possible values, which enables the model to capture the relevant sources of platforms' market power

[8]For a comprehensive review of the contemporary literature on two-sided markets, see Rietveld and Schilling (2021).

in a flexible and robust manner.[9] For example, the exogenous initial market shares capture differences between operating systems in the diversity of the smartphone models that adopt each of them. In Section 5, I extend the model by explicitly incorporating this form of diversity and show that all main results continue to hold.

The rest of this paper proceeds as follows. Section 2 presents the baseline model. Section 3 describes the solution concept adopted in the model and presents the solution. Section 4 analyzes the effects of super apps and the relevant restrictions on the market equilibrium. Section 5 evaluates the robustness of the baseline model by discussing two relevant extensions: one that accounts for model diversity among smartphone manufacturers using the same operating system, and another that considers environments in which platforms' developer-side prices are rigid. Section 6 concludes. The omitted mathematical content, unless specified otherwise, is relegated to the appendices.

2 The Model

I model the smartphone platform market as a two-sided market with cross-side network effects[10] and two competing operating systems, iOS and Android, as platforms that compete for users in the market. The two platforms, namely platform 0 and platform 1, are at the respective endpoints of a Hotelling line $[0,1]$ that characterizes the market. Two groups of unit-mass users, consumers and developers, are respectively on the two sides of the market and uniformly distributed on the Hotelling line so that $x \sim U[0,1]$ represents consumers' horizontal preferences and $y \sim U[0,1]$ represents those of developers. Consumers always use at most one of the two platforms—that is, they always single-home—while developers may operate on both platforms—that is, they may multi-home.

At the beginning of the game, each consumer is a patron of a platform $b \in \{0,1\}$ and incurs a switching cost when switching away from that platform. $f \in (0,1)$ denotes the fraction of consumers who initially patronize platform 0—that is, platform 0's (consumer-side) initial market share—and $1-f$ denotes the fraction associated with platform 1. I capture platforms' consumer-side market power, which arises primarily from consumers' switching costs, through their initial market shares. A platform with a larger initial market share therefore possesses greater consumer-side market power. Because there are only two platforms in the market, a platform with an initial market share larger than 0.5 is strong—equivalently,

[9] An alternative approach is to derive the initial market shares endogenously, as the steady-state levels in a dynamic model. However, the relatively short history of the current smartphone platform market—about 20 years since the iPhone's 2007 release—suggests that the market may not have reached its steady state before the emergence of super apps.

[10] As noted in Section 1, I assume away same-side network effects in the model.

holding a dominant position—on the consumer side, whereas a platform with an initial market share below 0.5 is weak on the consumer side. Consumers' horizontal preferences are independent of the platforms they initially patronize.[11] I assume platforms' initial market shares to be exogenous.[12]

The game proceeds as follows.

1. Each platform $i \in \{0,1\}$ simultaneously sets prices p_i to all consumers (abbreviated as C) and q_i to all developers (abbreviated as D).[13]

2. Observing all four prices posted by the two platforms, each consumer located at $x \in [0,1]$ on the Hotelling line and initially patronizing $b \in \{0,1\}$ chooses whether to use platform 0 or platform 1, and each developer located at $y \in [0,1]$ on the Hotelling line chooses whether to operate on platform 0, operate on platform 1, or multi-home.

3. Then, the game ends with all parties' payoffs realized as follows.

   - Denote consumers' baseline payoff from using platform $i \in \{0,1\}$ by $u_i > 0$; consumers' Hotelling transportation cost parameter by $s > 0$;[14] consumers' cross-side network effect parameter by $c > 0$; the number of developers that only operate on platform $i \in \{0,1\}$ by n_i^D; the number of multi-homing developers by n_2^D; and each consumer's switching cost by $k \geq 0$. This switching cost is symmetric—a consumer incurs the same cost k regardless of the direction of the switch. It captures the direct friction of switching rather than platform payoff differences, which are accounted for by baseline payoffs and horizontal preference heterogeneity.

     Then, for a consumer with $(x,b) \in [0,1] \times \{0,1\}$ using platform $i \in \{0,1\}$, the payoff is $U_i(x,b)$, where

     $$\begin{aligned} U_0(x,b) &= u_0 - sx + c(n_0^D + n_2^D) - p_0 - k\mathbb{1}[b=1], \\ U_1(x,b) &= u_1 - s(1-x) + c(n_1^D + n_2^D) - p_1 - k\mathbb{1}[b=0]. \end{aligned} \tag{2.1}$$

   - Denote developers' baseline payoff from operating on platform $i \in \{0,1\}$ by $v_i > 0$; developers' Hotelling transportation cost parameter by $t > 0$; developers' cross-side network effect parameter by $d > 0$; the number of consumers that use platform

[11] In other words, b is a random variable independent of x—that is, $b \perp x$—and with $P(b=0) = f$.

[12] As discussed in Section 1, the relatively short history of the current smartphone platform market justifies this assumption.

[13] In particular, platforms cannot price-discriminate.

[14] Consumers' horizontally differentiated preferences are associated with distinct platform features. For instance, Android offers extensive user customization, whereas iOS prioritizes standardization and limits platform flexibility.

$i \in \{0,1\}$ by n_i^C; and the amount of each multi-homer's cost saving by $r \geq 0$. For a developer, multi-homing always costs more than operating on a single platform. However, it costs less than developing for two platforms completely independently. Therefore, for each multi-homer, the cost saving r is non-negative. In addition, this cost saving is also "symmetric" in the sense that a multi-homer always enjoys the same amount of cost saving r, regardless of whether it has been operating on platform 0 and is new to platform 1, has been operating on platform 1 and is new to platform 0, or it is new to both platforms—developers are not assumed to initially patronize any platform.

Then, for a single-homing developer with $y \in [0,1]$ operating on platform $i \in \{0,1\}$, the payoff is $V_i(y)$, with

$$\begin{aligned} V_0(y) &= v_0 - ty + dn_0^C - q_0, \\ V_1(y) &= v_1 - t(1-y) + dn_1^C - q_1, \end{aligned} \tag{2.2}$$

and for a multi-homing developer with $y \in [0,1]$, the payoff is

$$V_2(y) = V_0(y) + V_1(y) + r. \tag{2.3}$$

- Let $m \geq 0$ denote the commission that each platform charges each developer operating on it for each consumer who uses it and is thus accessible to the developers. The commission m is non-negative, exogenous, and identical across platforms,[15] and it does not appear in users' payoff functions because the developer-side network effect d is defined net of the commission.

  Then, for each platform $i \in \{0,1\}$, its payoff is its profit, which is

$$\pi_i = p_i n_i^C + (q_i + m n_i^C)(n_i^D + n_2^D). \tag{2.4}$$

In this game, the strategy of each platform $i \in \{0,1\}$ is a pair of prices $(p_i, q_i) \in \mathbb{R}^2$ posted to consumers and developers, respectively. Platforms may set negative prices—that is, subsidize users. The strategy of a consumer is a function $L_C : [0,1] \times \{0,1\} \times \mathbb{R}^4 \to \{0,1\}$, where $L_C(x, b, p_0, q_0, p_1, q_1) = i \in \{0,1\}$ if and only if a consumer with $(x, b) \in [0,1] \times \{0,1\}$

[15] These assumptions are for the sake of both simplicity and consistency with real-world practice. In particular, the assumption that m is identical across platforms is mainly for simplicity. Assuming asymmetry leaves the model's results qualitatively unchanged but greatly complicates the associated mathematics. This symmetry assumption is also consistent with the fact that both Apple and Google have applied a standard 30% commission for a long period of time. As noted in Section 1, the recent changes in their respective commission rates are responses to antitrust lawsuits rather than to market forces.

chooses to use platform i given prices (p_0, q_0, p_1, q_1). The strategy of a developer is a function $L_D : [0,1] \times \mathbb{R}^4 \to \{0,1,2\}$, where $L_D(y, p_0, q_0, p_1, q_1) = i \in \{0,1,2\}$ if and only if a developer with $y \in [0,1]$ chooses to (1) operate on platform i only (when $i \in \{0,1\}$) or (2) multi-home (when $i = 2$) given prices (p_0, q_0, p_1, q_1). All else equal, each user prefers the platform closer to his or her location, and developers located near the midpoint of the Hotelling line are more likely to multi-home. Consumers' switching costs k discourage consumers from switching away from their initial platforms, while multi-homers' cost savings r encourage developers to multi-home.

Finally, among all of the terms defined above, the prices (p_0, q_0, p_1, q_1) and the market shares $(n_i^C)_{i=0}^1$ and $(n_i^D)_{i=0}^2$ are endogenous and jointly constitute the equilibrium of the game, while all other terms are exogenous.

3 Equilibrium of the Model

I use subgame perfect equilibrium as the solution concept for this game and refer to it as equilibrium thereafter. In the equilibrium, (1) given platforms' equilibrium prices and other users' equilibrium strategies, each user optimally chooses platform(s) to use or operate on to maximize his or her payoff, and (2) given users' equilibrium strategies and the other platform's equilibrium prices, each platform optimally sets its prices to maximize its profit.

Throughout this paper, I focus on equilibria characterized by interior solutions under full market coverage.[16] Such an equilibrium exists and is unique when (1) the degrees of users' horizontal differentiation are relatively large compared with the magnitudes of users' network effects, consumers' switching costs, and the differences in platforms' baseline payoffs on both sides of the market, (2) the platforms' baseline payoffs to consumers are relatively large compared with the switching costs and the degree of consumers' horizontal differentiation, and (3) the sum of platforms' baseline payoffs to developers, the magnitudes of users' network effects, the commission, and multi-homers' cost savings are all bounded above and below by different multiples of the degree of developers' horizontal differentiation.[17]

Assuming these conditions hold throughout the analysis, let $[(p_0^*, q_0^*), (p_1^*, q_1^*), L_C^*, L_D^*]$ denote the unique equilibrium of the model with market shares $(n_i^{C*})_{i=0}^1$ and $(n_i^{D*})_{i=0}^2$. The

[16] Specifically, the market is fully covered on both sides, neither platform serves all users on either side, and there are always some multi-homers. These resulting patterns are consistent with real-world outcomes in the smartphone platform market and in more general situations. This consistency justifies the focus.

[17] For the steps for solving the model and the relevant mathematical content, including the expressions associated with these conditions, see Appendix A.

equilibrium prices are

$$
\begin{aligned}
p_0^* &= \frac{4st - 2cd - (d+m)(v_+ + c + m)}{4t} \\
&\quad + \frac{[st(2c - d - m) - c^2 d]v_- + t[4st - d(3c+d) - m(c+2d) - m^2]u_-}{2t\phi}, \\
p_1^* &= \frac{4st - 2cd - (d+m)(v_+ + c + m)}{4t} \\
&\quad - \frac{[st(2c - d - m) - c^2 d]v_- + t[4st - d(3c+d) - m(c+2d) - m^2]u_-}{2t\phi}, \\
q_0^* &= \frac{v_+ - c - m}{4} + \frac{[3st - c(c+2d) - m(2c+d) - m^2]v_- + t(d - c - m)u_-}{2\phi}, \\
q_1^* &= \frac{v_+ - c - m}{4} - \frac{[3st - c(c+2d) - m(2c+d) - m^2]v_- + t(d - c - m)u_-}{2\phi},
\end{aligned}
\tag{3.1}
$$

where $\phi \coloneqq 6st - c^2 - d^2 - 4cd - 2m(c+d) - m^2$, $u_- \coloneqq u_0 - u_1 + k(2f-1)$, $v_- \coloneqq v_0 - v_1$, and $v_+ \coloneqq v_0 + v_1 + d + 2r$.

The equilibrium market shares are

$$
\begin{aligned}
n_0^{C*} &= \frac{1}{2} + \frac{(c+d+m)v_- + 2tu_-}{2\phi}, \\
n_1^{C*} &= \frac{1}{2} - \frac{(c+d+m)v_- + 2tu_-}{2\phi}, \\
n_0^{D*} &= 1 - \frac{v_+ + c + m}{4t} + \frac{(3st - cd)v_- + t(c+d+m)u_-}{2t\phi}, \\
n_1^{D*} &= 1 - \frac{v_+ + c + m}{4t} - \frac{(3st - cd)v_- + t(c+d+m)u_-}{2t\phi}, \\
n_2^{D*} &= \frac{v_+ + c + m}{2t} - 1.
\end{aligned}
\tag{3.2}
$$

The equilibrium strategy of each consumer is

$$
L_C^*(x, b, p_0, q_0, p_1, q_1) = \begin{cases} 0 & \text{if } x \le x_b^* \\ 1 & \text{elsewhere} \end{cases}, \tag{3.3}
$$

where $x_0^* = n_0^{C*} + \frac{k(1-f)}{s}$ is the location of the consumers who initially patronize platform 0 and are indifferent between the two platforms in the equilibrium, and $x_1^* = n_0^{C*} - \frac{kf}{s}$ is the location of the consumers who initially patronize platform 1 and are indifferent between the two platforms in the equilibrium.

The equilibrium strategy of each developer is

$$L_D^*(y, p_0, q_0, p_1, q_1) = \begin{cases} 0 & \text{if } y < n_0^{D*} \\ 1 & \text{if } y > 1 - n_1^{D*} \\ 2 & \text{elsewhere} \end{cases}, \tag{3.4}$$

where n_0^{D*} is the location of the developers who are indifferent between single-homing on platform 0 and multi-homing in the equilibrium, and $1 - n_1^{D*}$ is the location of the developers who are indifferent between single-homing on platform 1 and multi-homing in the equilibrium.

4 Comparative Statics

The heart of my analysis lies in the corollaries presented at the end of each part. These corollaries describe how restrictions on super apps affect each object of interest, thereby answering my research questions. However, because Apple's restrictions on super apps degrade super apps' functionalities and limit their cross-platform interoperability—therefore raising consumers' symmetric switching costs k and reducing multi-homers' cost savings r[18]—in each part of my comparative static analysis, I begin by examining how k and r affect each object of interest. Intuitively, each platform tends to prefer that consumers face lower costs when switching to it and higher costs when switching away, so that it can profit from serving more consumers. Nevertheless, the symmetry of k requires a closer examination of the relevant effects.

As a result, in each part of my comparative static analysis, the first few results are propositions that concern the effects of consumers' switching costs and multi-homers' cost savings. These propositions are mainly used to reach the corresponding corollaries, which concern the effects of restrictions on super apps.[19] Meanwhile, throughout the comparative static analysis in this paper, I hold all other exogenous variables[20] fixed and assume that the conditions in (A.1.9) and (A.2.1)–(A.2.4) are satisfied.

[18]Because both k and r are symmetric and the model abstracts from any direct utility associated with super apps, Apple's restrictions on super apps effectively impose the same restrictions on users of iOS's rival platforms as well.

[19]The proofs of Propositions 1 - 7 and Corollary 2 are relegated to Appendix B. The other corollaries follow directly from the relevant propositions. In particular, the proofs of Corollaries 3 and 4 closely resemble that of Corollary 2. Therefore, the proofs of these results are omitted from this paper for brevity.

[20]Such exogenous variables include platforms' baseline payoffs to users $(u_i, v_i)_{i=0}^1$, users' Hotelling transportation cost parameters s and t, users' network effect parameters c and d, the commission parameter m, and the initial market shares (f and $1 - f$).

4.1 Market Outcomes

I start my analysis by examining the equilibrium market outcomes—the prices in (3.1) and the market shares in (3.2).

Proposition 1 (Effects of Consumers' Switching Costs on Market Outcomes)**.** *When there is an increase in consumers' switching costs k, in equilibrium a platform that is strong (weak) on the consumer side*

1. *charges a higher (lower) price to consumers;*
2. *charges a higher (lower) price to developers if and only if the magnitude of the developer-side network effect is sufficiently large relative to that of the consumer-side network effect and the commission;*
3. *serves more (fewer) consumers; and*
4. *serves more (fewer) developers while serving the same number of multi-homers—that is, the number of multi-homers remains unchanged.*

Proposition 1 describes the effects of consumers' switching costs k on the equilibrium market outcomes. Part 1 concerns the effect on platforms' consumer-side prices. This effect is standard in the literature on switching costs and relates to the trade-off faced by the platforms. For each platform, the trade-off is between attracting more consumers from its rival by reducing the price and profiting more from each consumer using the platform by raising the price. When k increases, it becomes more difficult for each platform to attract additional consumers from its rival. Thus, a platform that is strong on the consumer side finds it more profitable to raise its consumer-side price because its relatively large consumer base—that is, its relatively large initial market share—provides a greater scope for exploitation. In contrast, a platform that is weak on the consumer side, despite the increased difficulty in attracting consumers, reduces its consumer-side price because it does not have a sufficiently large consumer base for profitable exploitation. As for the effect via cross-side network effects, when k increases, though a strong platform has some incentive to reduce its consumer-side price in order to positively affect the part of its profit that comes from the developer side, the profitability of exploiting large consumer base turns out to outweigh this incentive. In contrast, a weak platform reduces its consumer-side price to attract more consumers—though unsuccessfully (see part 3). When neither of the two platforms holds a dominant position on the consumer side—that is, when their initial market shares are equal ($f = 1 - f = 0.5$)—neither aspect of the above trade-off outweighs the other and, as a result, changes in k do not affect platforms' consumer-side prices.

Part 2 shows that the effect of k on the platforms' developer-side prices depends on a comparison that involves the magnitudes of the two groups of users' network effects and the commission. In particular,

$$\frac{\partial q_0^*}{\partial k} = -\frac{\partial q_1^*}{\partial k} = \frac{t(d-c-m)(2f-1)}{2\phi}. \tag{4.1.1}$$

Accordingly, the direction of the effect depends on (1) the platform's position on the consumer side, captured by the sign of $2f-1$, and (2) the relative magnitude of cross-side network effects and fees, captured by the sign of $d-c-m$. On the one hand, when d increases, the developer-side network effect becomes larger, making it more profitable for a platform to use its market power to exploit developers by charging them a higher price. This is the price dimension through which a platform may profit from adjusting its developer-side price. On the other hand, when c increases, the consumer-side network effect becomes larger. Then, a platform finds it more profitable to attract additional developers by reducing its developer-side price so as to further exploit consumers by using its market power. This is the quantity dimension through which a platform may profit from adjusting its developer-side price. The commission boosts the quantity dimension: when m increases, a platform finds it more profitable to attract more developers—and then more consumers, via the channel of the consumer-side network effect—so as to earn more from the commission.

The number of consumers served by a platform is related to its consumer-side market power. A platform that is strong on the consumer side has a larger consumer base to exploit and therefore possesses greater consumer-side market power than its rival. Thus, when k increases, such a strong platform would raise its developer-side price if and only if $d > c+m$—that is, the extractable surplus associated with the developer-side network effect is larger than that associated with the consumer-side network effect and the commission. If, instead, $d < c+m$, it would reduce its developer-side price to facilitate its extraction via the channel of the commission and exploitation of the consumer-side network effect.

As for a platform that is weak on the consumer side, its relevant reaction is exactly the opposite of that of a strong one. A weak platform avoids competing with its strong rival in the same dimension and shifts to the dimension less profitable for the rival. For example, when $d > c+m$, the strong rival finds it more profitable to raise its developer-side price in order to extract surplus from developers, while the weak platform correspondingly reduces its developer-side price to profit through the quantity dimension. Similar to the situation in part 1, the effect in this part disappears when $f = 0.5$—that is, when no platform holds a dominant position on the consumer side. It also disappears when $d = c+m$—that is, neither of the above two dimensions outweighs the other.

Part 3 and part 4 reflect how a platform's market power influences the effect of k on the equilibrium market shares. When k increases, consumers become less willing to switch. Thus, a platform that is strong on the consumer side ends up serving a larger number of consumers, even though it raises its consumer-side price to extract more surplus from them. Meanwhile, this increase in its consumer-side market share increases the amount of network effect that the developers operating on it can enjoy, enabling it to attract more developers. The increase in its developer-side market share can further increase its consumer-side market share via the channel of the consumer-side network effect and, subsequently, its profitability from raising its consumer-side price, forming a virtuous cycle for it. Moreover, the number of multi-homers remains invariant to changes in consumers' switching costs. This is because an increase in k generates exactly offsetting flows: it pushes as many single-homers who operate on the weak platform into multi-homing as it induces multi-homers to operate exclusively on the strong platform. A weak platform faces the opposite situation. In addition, as in the previous parts, when neither platform is strong on the consumer side, the equilibrium market shares are invariant to any change in k.

Proposition 2 (Effects of Multi-homers' Cost Savings on Market Outcomes)**.** *When there is an increase in multi-homers' cost savings r, in equilibrium each of the platforms*

1. *charges a lower price to consumers;*
2. *charges a higher price to developers;*
3. *still serves the same number of consumers; and*
4. *serves more developers, with some developers switching from single-homing to multi-homing.*

Proposition 2 is about the effects of multi-homers' cost savings r on the equilibrium market outcomes. In contrast to consumers' switching costs k, the effects of r are symmetric in that they are invariant to platforms' consumer-side market power. Part 1 implies that when r increases, both platforms charge consumers lower prices. This is because their higher developer-side prices—see part 2—respectively incentivize them to charge consumers lower prices in order to attract more developers via the channel of the developer-side network effect.

Part 2 shows the direct effect of an increase in r, which is on the developer side. When r increases, multi-homing becomes more attractive for each developer. This makes it profitable for both platforms to charge developers higher prices.

Part 3 closely reflects the symmetry of the effects of r. Because r affects both the platforms' consumer-side prices and their developer-side market shares in a symmetric manner,

combined with (2.1), each consumer's choice between the platforms is unaffected, leaving consumer-side market shares unaffected as well.

Part 4 indicates that the increases in platforms' developer-side prices induce some single-homers on each platform to multi-home. From (2.2) and (2.3), when r increases, each developer's payoff from multi-homing is unaffected, whereas the payoff from single-homing decreases regardless of which platform the developer operates on.[21] Thus, when r increases, on each platform there are some developers who switch from single-homing to multi-homing.

Because restrictions on super apps affect the market by raising consumers' switching costs and reducing multi-homers' cost savings, Propositions 1 and 2 together yield the following corollary.

Corollary 1 (Effects of Restrictions on Super Apps on Market Outcomes)**.** *In response to restrictions on super apps,*

1. *a platform that is strong on the consumer side charges a higher price to consumers and ends up serving more consumers;*

2. *a platform that is weak on the consumer side charges a lower price to developers and ends up serving fewer users on both sides;*

3. *the platforms end up serving fewer multi-homers; and*

4. *the direction of change in each of the other market outcomes is ambiguous.*

Corollary 1 summarizes the effects of restrictions on super apps on the equilibrium market outcomes. Each of the effects presented in the preceding three parts is unambiguous because the effects of the restrictions operate in the same direction through the two channels—consumers' switching costs k and multi-homers' cost savings r. For example, both an increase in k and a decrease in r result in a higher consumer-side price of a strong platform. Therefore, in response to restrictions on super apps, which both raise consumers' switching costs and reduce multi-homers' cost savings, a strong platform charges a higher price to consumers.

In terms of part 4, for each of the other market outcomes, the effects of the restrictions operate in opposite directions through the two channels, leaving the overall effect ambiguous. The exact direction of this overall effect depends critically on the extent to which the restrictions affect the values of k and r. For example, in response to restrictions on super

[21] In terms of multi-homing, the positive effect via the channel of r exactly offsets the negative effect via the channel of q_0 and q_1 ($\frac{\partial q_0^*}{\partial r} = \frac{\partial q_1^*}{\partial r} = \frac{1}{2}$), whereas for single-homing on platform $i \in \{0,1\}$, r and q_{1-i} are absent from the payoff, leaving the negative effect via q_i alone.

apps, if the magnitude of the increase in k is sufficiently larger than that of the decrease in r—so that the positive effect via the channel of k outweighs the negative effect via the channel of r—a strong platform would end up serving more developers. Otherwise, a strong platform would end up serving fewer developers in response to these restrictions.

4.2 Profits and Welfare

First, from (2.4), the equilibrium profit of platform $i \in \{0, 1\}$ is

$$\pi_i^* = p_i^* n_i^{C*} + (q_i^* + m n_i^{C*})(n_i^{D*} + n_2^{D*}). \tag{4.2.1}$$

Proposition 3 (Effects of Consumers' Switching Costs on Profits)**.** *An increase in consumers' switching costs k increases (decreases) the equilibrium profit of a platform if and only if the platform is strong (weak) on the consumer side.*

Proposition 3 illustrates the effects of consumers' switching costs k on platforms' equilibrium profits. When k increases, a platform is better off if and only if it is strong on the consumer side. From Proposition 1, for a strong platform, an increase in k always increases its market shares on both sides of the market and its consumer-side price, and its developer-side price would decrease if and only if it finds such a price reduction to be profitable. As a result, an increase in k benefits it via the channel of both prices (i.e., intensively) and market shares (i.e., extensively, via quantities).

In contrast, a weak platform becomes worse off when k increases. This increase leads to decreases in both its consumer-side price and its market shares on both sides of the market. Although it may raise its developer-side price in this situation, its lack of consumer-side market power means that any resulting gain is relatively small and cannot completely offset its losses from the relevant decreases. When neither of the platforms holds a dominant position on the consumer side, again from Proposition 1, prices and market shares are invariant to any change in k. Therefore, from (4.2.1), platforms' profits are invariant to any change in k as well.

Proposition 4 (Effects of Multi-homers' Cost Savings on Profits)**.** *An increase in multi-homers' cost savings r increases the equilibrium profit of each platform.*

Proposition 4 is about the effects of multi-homers' cost savings r on platforms' equilibrium profits. It implies that an increase in r increases both platforms' equilibrium profits. From Proposition 2, for each platform, both its developer-side price and the number of developers it serves increase in r. The positive effect on equilibrium profits through these two dimensions outweighs the negative effect through the consumer-side price reductions.

Propositions 3 and 4 together characterize platforms' attitudes toward super apps and the relevant restrictions, yielding the following corollary.

Corollary 2 (Effects of Restrictions on Super Apps on Profits)**.** *In response to restrictions on super apps,*

1. *the equilibrium profit of a platform that is not strong on the consumer side decreases; and*

2. *the direction of change in the equilibrium profit of a platform that is strong on the consumer side is ambiguous: the profit increases (decreases) when the restrictions increase consumers' switching costs to a significantly greater (lesser) extent than they decrease multi-homers' cost savings.*

Corollary 2 presents the effects of restrictions on super apps on platforms' equilibrium profits. Because restrictions on super apps affect the market by raising consumers' switching costs k and reducing multi-homers' cost savings r, such restrictions always reduce a platform's equilibrium profit via the channel of r, and they increase its profit via the channel of k if and only if the platform is strong on the consumer side. Accordingly, part 1 suggests that restrictions on super apps make a platform that is not strong on the consumer side worse off. As a result, such a platform does not prefer any restriction on super apps.

In contrast, part 2 shows that a strong platform faces a trade-off between the effects operating through the two channels and may therefore be better off under such restrictions. Similar to the situation in Corollary 1, the preference of a strong platform depends on the extent to which the restrictions affect the values of k and r. In particular, for a strong platform, because its equilibrium profit increases in r, it would find restrictions on super apps preferable if the resulting increase in k is sufficiently large—compared with the decrease in r—to protect it effectively from consumer-side price competition. Similarly, when the increase in k is relatively small compared with the decrease in r, a strong platform does not find restrictions on super apps desirable. For example, the numerical simulation results shown in Figure 1 present scenarios (black region) where restrictions on super apps—which increase consumers' switching costs by Δk and decrease multi-homers' cost savings by Δr—increase the equilibrium profit of a strong platform. These scenarios are such that Δk, the increase in k, is sufficiently large compared with Δr, the decrease in r.[22] Accordingly, any decision by a strong platform to restrict super apps reveals that it expects parameter conditions to lie within the black region, rather than the gray region, of Figure 1.

[22]Figure 1 was generated using MATLAB. The parameter values used to generate Figure 1 are as follows: $s \in [2.1, 2.13]$, $t = 2.1$, $c = d = 1.999$, $m = 0.2$, $u_0 = u_1 = v_0 = v_1 = 1$, $k = 0.63$, $r = 0.03$, and $f = 0.58$—accordingly, platform 0 is strong in the market. These values all satisfy (A.1.9) and (A.2.1)–(A.2.4).

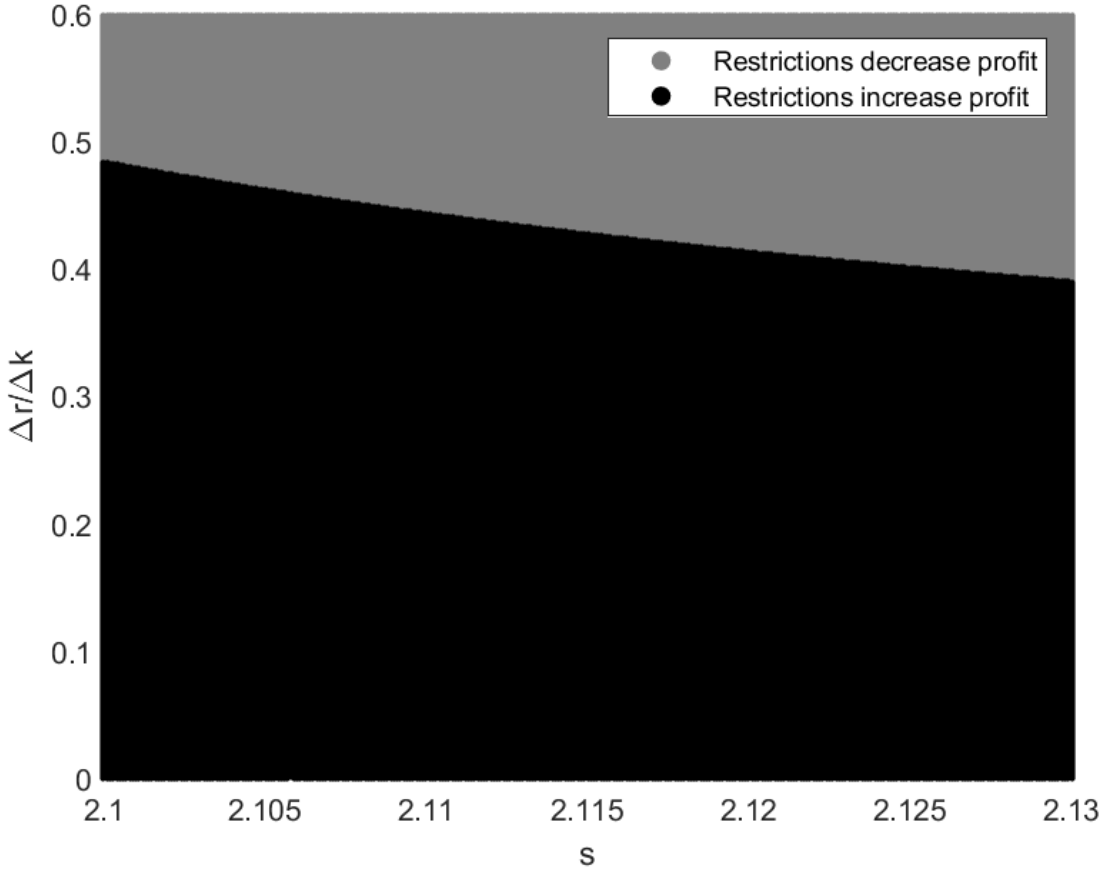


Figure 1: Direction of the Effect of Super App Restrictions on Strong Platform Profit, Degree of Consumers' Horizontal Differentiation $s \in [2.1, 2.13]$

Next, by the nature of the two-sided market, I analyze the equilibrium surplus of the consumers and that of the developers separately. The equilibrium consumer surplus is

$$\begin{aligned} CS^* &= f\Big[\int_0^{x_0^*} U_0^*(x,0)dx + \int_{x_0^*}^1 U_1^*(x,0)dx\Big] + (1-f)\Big[\int_0^{x_1^*} U_0^*(x,1)dx + \int_{x_1^*}^1 U_1^*(x,1)dx\Big] \\ &= u_1 + c(n_1^{D*} + n_2^{D*}) - p_1^* + sn_0^{C*2} - fk + \frac{k^2 f(1-f)}{s} - \frac{s}{2}, \end{aligned} \tag{4.2.2}$$

where $U_i^*(x,b)$ is calculated in accordance with (2.1) and denotes the equilibrium payoff of a consumer with $x \in [0,1]$, initially patronizing platform $b \in \{0,1\}$, and using platform

$i \in \{0,1\}$. The equilibrium developer surplus is

$$\begin{aligned} DS^* &= \int_0^{n_0^{D*}} V_0^*(y)dy + \int_{n_0^{D*}}^{n_0^{D*}+n_2^{D*}} V_2^*(y)dy + \int_{n_0^{D*}+n_2^{D*}}^{1} V_1^*(y)dy \\ &= \frac{t}{2}[(n_0^{D*}+n_2^{D*})^2 + (n_1^{D*}+n_2^{D*})^2] - r, \end{aligned} \tag{4.2.3}$$

where $V_i^*(y)$ is calculated in accordance with (2.2) and (2.3), and it denotes the equilibrium payoff of a developer with $y \in [0,1]$ who operates on platform i (when $i \in \{0,1\}$) or multi-homes (when $i = 2$).

Proposition 5 (Effects of Consumers' Switching Costs on Consumer Surplus)**.** *When there is an increase in consumers' switching costs k, the equilibrium consumer surplus*

1. *decreases when the degree of consumers' horizontal differentiation is sufficiently large relative to consumers' switching costs and the magnitudes of users' network effects;*

2. *may increase when the degree of consumers' horizontal differentiation is not sufficiently large.*

Proposition 5 characterizes how an increase in consumers' switching costs k affects the equilibrium consumer surplus. Part 1 indicates that the equilibrium consumer surplus decreases in k as long as consumers are sufficiently horizontally differentiated—specifically, when s, the degree of consumers' horizontal differentiation, is sufficiently large relative to k and to the network effect parameters c and d.[23] When consumers are sufficiently horizontally differentiated, many still opt to switch despite rising costs in order to reach their preferred platform. More formally, when s is sufficiently large relative to c and d, each consumer's payoff from using a (weakly) strong platform decreases in k regardless of which platform the consumer initially patronizes. When k increases, though the number of developers serving on a strong platform rises, the platform's consumer-side price also increases, outweighing consumers' additional network-effect benefit.[24] Meanwhile, for the consumers who initially

[23] Mathematically, it requires $s > 2k$ and $s > \frac{c^2+d^2+4cd+2m(c+d)+m^2}{4t}$. A substantial degree of consumers' horizontal differentiation ensures sufficient heterogeneity for a significant range of consumers to be willing to switch to the platforms that are respectively closer to them, and an increase in k adversely affects consumers via this channel. These conditions are qualitatively aligned with the conditions on large degrees of users' horizontal differentiation stated in Section 3. Quantitatively, however, neither set of conditions implies the other.

[24] In particular, assume without loss of generality that $f \in [0.5,1)$ so that platform 0 is (weakly) strong on the consumer side. Then, combining $\frac{\partial p_0^*}{\partial k} = -\frac{\partial p_1^*}{\partial k} = \frac{[4st-d(3c+d)-m(c+2d)-m^2](2f-1)}{2\phi}$, $\frac{\partial n_0^{D*}}{\partial k} = -\frac{\partial n_1^{D*}}{\partial k} = \frac{(c+d+m)(2f-1)}{2\phi}$, and $\frac{\partial n_2^{D*}}{\partial k} = 0$ yields $\frac{\partial U_0^*(x,b)}{\partial k} = -\frac{(2f-1)(\phi-2st)}{2\phi} - \mathbb{1}[b=1]$ and $\frac{\partial U_1^*(x,b)}{\partial k} = \frac{(2f-1)(\phi-2st)}{2\phi} - \mathbb{1}[b=0]$. Since $s > \frac{c^2+d^2+4cd+2m(c+d)+m^2}{4t}$ is equivalent to $\phi > 2st$, and because $f \in [0.5,1)$, it follows that for any $x \in [0,1]$, $\frac{\partial U_0^*(x,b)}{\partial k} < 0$ for each $b \in \{0,1\}$.

patronize a (weakly) weak platform, their payoffs from switching to the strong platform always decrease in k. The only exception is the consumers who initially patronize the weak platform and choose to continue using it: both their payoffs and their number increase in k (because the price reduction outweighs the decrease in the number of developers). As long as k is small relative to s, however, the number of them is relatively small, and the overall effect of an increase in k on the equilibrium consumer surplus is negative.[25] Consequently, a high value of s relative to c, d, and k both makes the payoffs of most groups of consumers decrease in k and reduces the size of the group whose payoffs increase in k. As a result, the overall effect of an increase in consumers' switching costs on the equilibrium consumer surplus is negative in this situation.

In contrast, part 2 suggests that the equilibrium consumer surplus may increase in k when consumers are not very horizontally differentiated. This happens, for instance, when a sufficiently large number of consumers and developers cluster on the stronger platform, generating substantial cross-side network effects. In detail, when s is relatively small compared with c and d, and one of the two platforms is sufficiently stronger than the other, each consumer's payoff from using the strong platform increases in k, regardless of which platform the consumer initially patronizes.[26] In this situation, the strong platform's consumer-side price may even decrease in k.[27] As a result, when k increases, consumers who use this platform may benefit both from this price reduction and from the resulting increase in the platform's equilibrium developer-side market share, which enlarges the network-effect gain available on the platform. Thus, the positive effect of an increase in k on the payoffs of consumers who use the strong platform may outweigh the negative effect of the increase on the payoffs of consumers who use the weak platform,[28] and the overall effect of an increase in consumers' switching costs on the equilibrium consumer surplus may therefore be positive. For example, the numerical simulation results shown in Figure 2 illustrate that the equilibrium consumer surplus tends to decrease in k when s is relatively large but may increase in k when s is sufficiently small.[29]

[25] Still assume without loss of generality that $f \in [0.5, 1)$ so that platform 1 is (weakly) weak on the consumer side. Then, for payoffs, combined with $\phi > 2st$, for any $x \in [0,1]$, $\frac{\partial U_1^*(x,0)}{\partial k} < \frac{\phi - 2st}{2\phi} - 1 < 0$ and $\frac{\partial U_1^*(x,1)}{\partial k} = \frac{(2f-1)(\phi-2st)}{2\phi} \geq 0$. As for the number of consumers, combining the definition of f with (3.3), the size of this group of consumers is $(1-f)(1-x_1^*) = (1-f)(n_1^{C*} + \frac{kf}{s})$, whose derivative with respect to k is $(1-f)(\frac{f}{s} + \frac{\partial n_1^{C*}}{\partial k}) = (1-f)[\frac{f}{s} - \frac{t(2f-1)}{\phi}] = \frac{st + f(\phi - 2st)}{s\phi} > 0$.

[26] For example, let $s \leq \frac{4[c^2+d^2+4cd+2m(c+d)+m^2]}{23t}$ so that $\phi \leq \frac{st}{4}$ and $f = 0.75$, then for each $b \in \{0,1\}$, $\frac{\partial U_0^*(x,b)}{\partial k} = -\frac{(2f-1)(\phi-2st)}{2\phi} - \mathbb{1}[b=1] \geq -\frac{0.5(\phi-8\phi)}{2\phi} - \mathbb{1}[b=1] = 1.75 - \mathbb{1}[b=1] > 0$.

[27] With $f \in [0.5, 1)$, $\frac{\partial p_0^*}{\partial k} = \frac{[4st - d(3c+d) - m(c+2d) - m^2](2f-1)}{2\phi} \leq 0$ if $s \leq \frac{c^2+d^2+4cd+2m(c+d)+m^2}{4t}$.

[28] With $f \in [0.5, 1)$, for each $b \in \{0,1\}$, $\frac{\partial U_1^*(x,b)}{\partial k} = \frac{(2f-1)(\phi-2st)}{2\phi} - \mathbb{1}[b=0] \leq 0$ if $s \leq \frac{c^2+d^2+4cd+2m(c+d)+m^2}{4t}$.

[29] Figure 2 was generated using MATLAB. The parameter values used to generate Figure 2 are identical

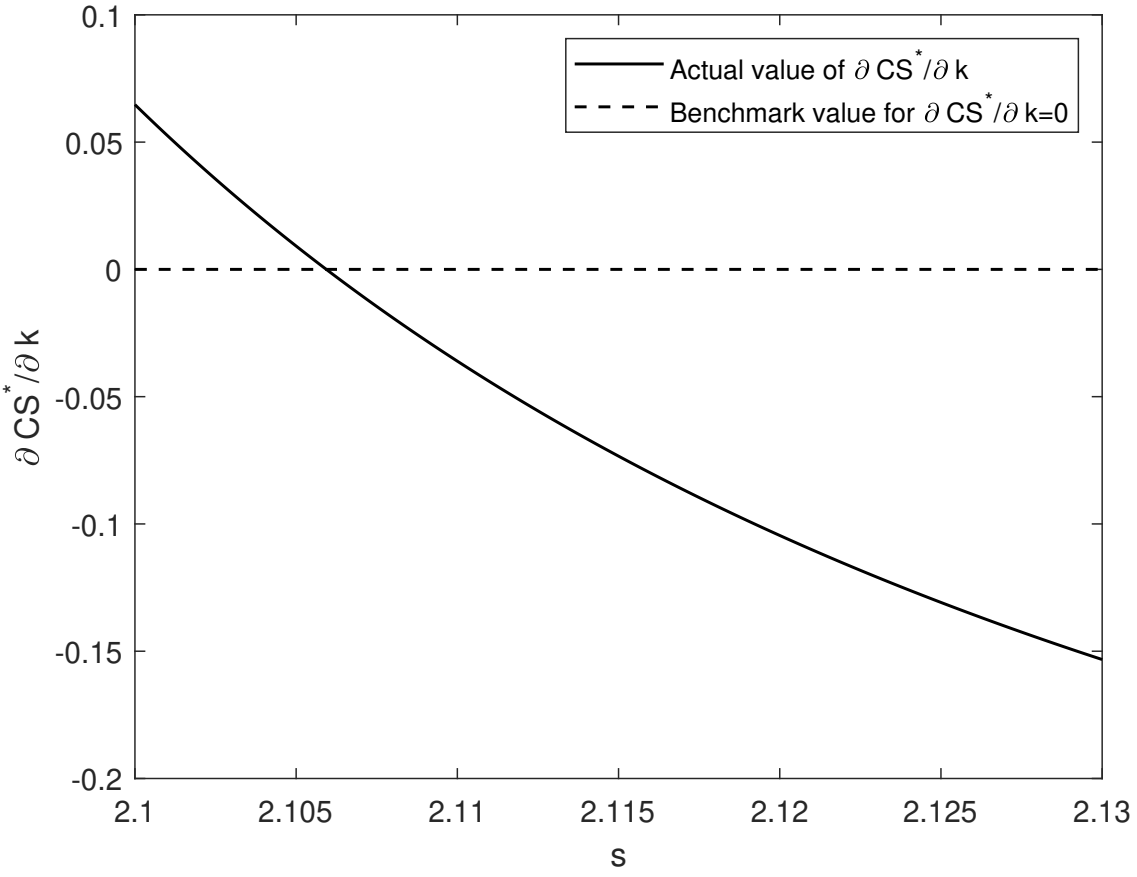


Figure 2: Effect of Consumers' Switching Costs k on Consumer Surplus, Degree of Consumers' Horizontal Differentiation $s \in [2.1, 2.13]$

Proposition 6 (Effects of Consumers' Switching Costs on Developer Surplus)**.** *When there is an increase in consumers' switching costs k,*

1. *the equilibrium developer surplus increases if and only if the strong platform serves more developers in equilibrium than the weak platform; and*

2. *the resulting change in the equilibrium developer surplus diminishes in magnitude and ultimately vanishes as consumers become sufficiently horizontally differentiated.*

Proposition 6 describes the effect of changes in consumers' switching costs k on the equilibrium developer surplus. Part 1 identifies the key factor that determines the direction of the effect: the contrast between the two platforms' equilibrium developer-side market shares. From (2.2) and (2.3), consumers' switching costs affect each developer's payoff via two channels: consumer-side market shares and developer-side prices. For the payoff from

to those in Figure 1.

operating on a strong platform, the positive effect of an increase in k via the resulting increase in the equilibrium consumer-side market share outweighs the negative effect via the resulting increase in the consumer-side price, and the directions of these effects are reversed for a weak platform. Thus, when k increases, each single-homer's payoff from operating on a strong platform increases, the payoff from operating on a weak platform decreases, and each multi-homer's payoff remains unchanged.[30] Therefore, the positive effect of an increase in k on the payoffs of developers operating on the strong platform outweighs the negative effect on those operating on the weak platform if and only if the strong platform serves more developers than the weak platform.

Part 2 suggests that the effect of k on the equilibrium developer surplus is negligible when consumers are sufficiently horizontally differentiated. As consumers become sufficiently horizontally differentiated, the effects of consumers' switching costs operating through both consumer-side market shares and developer-side prices diminish in magnitude and ultimately vanish. Consequently, the overall effect on each developer's payoff—and hence on the equilibrium developer surplus—exhibits the same pattern.[31]

Proposition 7 (Effects of Multi-homers' Cost Savings on Welfare)**.** *An increase in multi-homers' cost savings r*

1. *increases the equilibrium consumer surplus; and*

2. *decreases the equilibrium developer surplus.*

Proposition 7 is about the effects of multi-homers' cost savings r on the equilibrium surpluses. Part 1 indicates that the equilibrium consumer surplus increases in r. The explanation for this comes directly from Proposition 2, whose part 1 shows that when r increases, all consumers pay lower prices regardless of the platforms they respectively use; part 3, together with (3.3), implies that no consumer chooses a different platform when there is any change in r; and part 4 indicates that every consumer enjoys a larger amount of network effect when r increases. As a result, when r increases, each consumer pays a lower price and enjoys a larger amount of network effect, leading to an increase in the equilibrium consumer surplus.

Part 2 says that the equilibrium developer surplus decreases in r. This result is quite surprising because such an increase seems to be directly beneficial to developers. The reason

[30] In particular, combining $f \in [0.5, 1)$, (2.2), (2.3), $\frac{\partial n_0^{C*}}{\partial k} = -\frac{\partial n_1^{C*}}{\partial k} = \frac{t(2f-1)}{\phi}$, and $\frac{\partial q_0^*}{\partial k} = -\frac{\partial q_1^*}{\partial k} = \frac{t(d-c-m)(2f-1)}{2\phi}$ yields $\frac{\partial V_0^*(y)}{\partial k} = -\frac{\partial V_1^*(y)}{\partial k} = \frac{t(d+c+m)(2f-1)}{2\phi} \geq 0$ and $\frac{\partial V_2^*(y)}{\partial k} = 0$.

[31] For $\frac{\partial n_0^{C*}}{\partial k} = -\frac{\partial n_1^{C*}}{\partial k} = \frac{t(2f-1)}{\phi}$, $\frac{\partial q_0^*}{\partial k} = -\frac{\partial q_1^*}{\partial k} = \frac{t(d-c-m)(2f-1)}{2\phi}$, $\frac{\partial V_0^*(y)}{\partial k} = -\frac{\partial V_1^*(y)}{\partial k} = \frac{t(d+c+m)(2f-1)}{2\phi}$, and $\frac{\partial V_2^*(y)}{\partial k} = 0$, since $\phi = 6st - c^2 - d^2 - 4cd - 2m(c+d) - m^2$, all of these values converge to 0 as $s \to +\infty$.

for this result lies, again, in Proposition 2, whose part 2 points to the main driving force of this surprising result: when r increases, both platforms proportionally raise their developer-side prices. Thus, because the developer-side network effect remains unaffected (by part 3 of Proposition 2), an increase in r decreases every developer's payoff from single-homing and does not affect his or her payoff from multi-homing.[32] Therefore, the equilibrium developer surplus decreases in r.

Propositions 5 – 7 jointly yield the welfare implications associated with super apps and the relevant restrictions. When s is small—that is, when consumers are not very horizontally differentiated so that sufficiently many of them are strongly motivated to switch platforms—the effects of restrictions on super apps on users are ambiguous. In contrast, when consumers are sufficiently horizontally differentiated—that is, when s is large—the three propositions yield some definitive results.

Corollary 3 (Effects of Restrictions on Super Apps on Welfare)**.** *In response to restrictions on super apps,*

1. *when consumers are sufficiently horizontally differentiated, the equilibrium consumer surplus decreases, while the equilibrium developer surplus increases;*

2. *when consumers are not very horizontally differentiated and the restrictions increase consumers' switching costs to a significantly greater extent than they decrease multi-homers' cost savings, the equilibrium consumer surplus may increase, and the equilibrium developer surplus may decrease.*

Corollary 3 directly follows Propositions 5 – 7. Part 1 illustrates how restrictions on super apps affect users when consumers are sufficiently horizontally differentiated. In this situation, super apps benefit consumers through both channels. Developers, however, are worse off with super apps, because the effect on the equilibrium developer surplus operating through the channel of multi-homers' cost savings outweighs the effect operating through consumers' switching costs. This is because the latter effect is relatively small in magnitude when s is sufficiently large—it vanishes as s goes to infinity. Consequently, consumers are worse off under any restriction on super apps, while developers are better off, as such restrictions reduce multi-homers' cost savings and thereby limit platforms' ability to extract developer surplus.

[32] This explanation is consistent with the quantitative results that $\frac{\partial DS^\star}{\partial r} = -\frac{1-n_2^{D\star}}{2}$ and $\frac{\partial q_0^\star}{\partial r} = \frac{\partial q_1^\star}{\partial r} = \frac{1}{2}$: the fraction of single-homers is $1-n_2^{D*}$, and when r increases, the resulting higher developer-side prices adversely affect all single-homers.

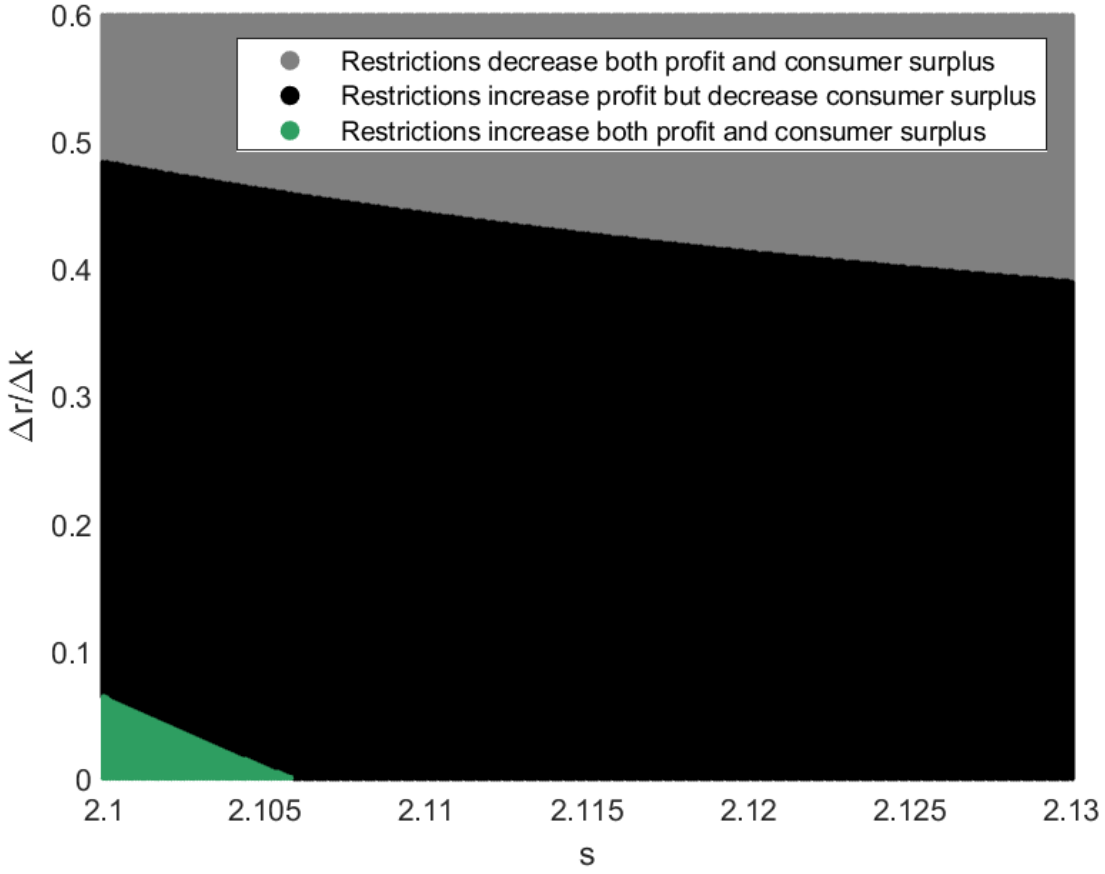


Figure 3: Direction of the Effect of Super App Restrictions on Strong Platform Profit and Consumer Surplus, Degree of Consumers' Horizontal Differentiation $s \in [2.1, 2.13]$

Part 2 qualitatively describes the ambiguity in the effects of restrictions on super apps on users when consumers are not very horizontally differentiated. In this situation, the equilibrium consumer surplus increases in both consumers' switching costs and multi-homers' cost savings, whereas the equilibrium developer surplus decreases in both of them when the strong platform serves fewer developers in the equilibrium than the weak platform. Because restrictions on super apps increase consumers' switching costs and reduce multi-homers' cost savings, the effects operating through consumers' switching costs and those operating through multi-homers' cost savings may move in opposite directions for both equilibrium consumer surplus and equilibrium developer surplus. Thus, when the restrictions increase consumers' switching costs to a significantly greater extent than they decrease multi-homers' cost savings, the effects operating through switching costs may outweigh those operating through cost savings. For example, the numerical simulation results shown in Figure 3 resent scenarios (green region) where restrictions on super apps increase the equilibrium consumer

surplus in addition to increasing the equilibrium profit of a strong platform. These scenarios are such that (1) Δk, the resulting increase in consumers' switching costs, is sufficiently large compared to Δr, the resulting decrease in multi-homers' cost savings, and (2) s is sufficiently small.[33] As a result, in this situation, restrictions on super apps may positively affect the equilibrium consumer surplus. Similarly, they may negatively affect the equilibrium developer surplus when the strong platform serves fewer developers in the equilibrium than the weak platform, so that the effect operating through consumers' switching costs is negative.

Corollaries 2 and 3 establish the relationship between platforms' consumer-side market power and their attitudes toward restrictions on super apps, as well as the relevant antitrust-related implications of these restrictions. I discuss these issues in Section 4.3.

4.3 The U.S.–Asia Difference in Apple's Attitude toward Restrictions on Super Apps and the Antitrust-Related Implications of These Restrictions

Corollary 2 provides an explanation for the regional difference in Apple's attitude toward super apps and the relevant restrictions. In Asia, the market share of iOS is lower than that of Android. Thus, iOS is weak on the consumer side and, therefore, benefits from super apps via the channels of both consumers' switching costs and multi-homers' cost savings. In Asia, Apple's restrictions on super apps are largely absent, as my analysis predicts. However, iOS holds a dominant position on the consumer side of the U.S. smartphone platform market. Hence, in the United States, Apple may find it profitable to restrict super apps, and it has indeed imposed a great variety of restrictions on super apps, implying that it has considered such restrictions to be profitable. From part 2 of Corollary 2, such restrictions are profitable when Apple finds that the resulting increase in consumers' switching costs is sufficiently large to protect it effectively from consumer-side price competition, so that this positive effect outweighs the negative effect of such restrictions operating through the decrease in multi-homers' cost savings.

The analysis of Apple's contrasting behavior in Asia and the U.S. also has implications for its potential antitrust concerns in the United States. From part 2 of Corollary 2, since iOS is strong in the United States, Apple has an incentive to restrict super apps only when the restrictions increase consumers' switching costs to a significantly greater extent than they decrease multi-homers' cost savings. In this situation, however, part 2 of Corollary 3 shows that consumers may also benefit from restrictions on super apps when platforms

[33] Figure 3 was produced using MATLAB. The numerical parameter values used in the simulation are the same as those used for Figure 2.

are sufficiently close substitutes. For instance, Figure 3 implies that when Apple's revealed preference is to restrict super apps, the effect of the restrictions on consumer welfare remains ambiguous. On the other hand, part 1 of Corollary 3 shows that when consumers perceive the platforms as highly differentiated, the restrictions adversely affect them. In particular, under this condition, they are worse off under any restriction on super apps via both of the above channels. The effects via the two channels operate in the same direction, thus reinforcing one another. Thus, whether Apple's restrictions adversely affect consumers depends on whether consumers view platforms as highly differentiated or close substitutes and the extent to which the restrictions affect consumers' switching costs and multi-homers' cost savings—no definitive conclusion can be reached without these two pieces of information.

In contrast, even when consumers perceive the platforms as highly differentiated, the restrictions' effects via the two channels on developers are not necessarily in the same direction. In this situation, only the effect via the channel of multi-homers' cost savings is substantial, and it is positive. Thus, developers are better off with the decrease in multi-homers' cost savings resulting from the restrictions. Therefore, the above results imply that in the relevant antitrust-related analyses, the effects of such restrictions on consumers should be regarded as first-order in importance relative to those on developers.

5 Discussion

The results and implications presented in Section 4 remain robust across all parameter values that satisfy (A.1.9) and (A.2.1)–(A.2.4). However, several additional influencing factors lie outside the scope of the baseline model. One of these factors is platforms' degrees of internal diversity, that is, the model diversity among manufacturers using the same operating system. The internal diversity of iOS and Android in the smartphone platform market need not be the same across regions,[34] and the more internally diversified a platform is, the more users it could easily attract. Thus, I extend the baseline model by letting the two platforms be respectively located at l_0 and l_1 on the Hotelling line with $0 \leq l_0 < l_1 \leq 1$ to investigate the effects of platforms' internal diversity. The exogenous initial market shares in the baseline model implicitly capture this factor, whereas this extension incorporates it explicitly. I find that under this extension, all main results presented in Section 4 are still valid as long as an equilibrium exists.[35]

[34]For example, many Android manufacturers sell much wider ranges of models in Asia than they do in the United States. Representative examples of such manufacturers include Huawei and Xiaomi in China. Several factors contribute to this phenomenon. In addition to political reasons, another relevant factor concerns carriers: few U.S. carriers officially support a wide range of Huawei and Xiaomi models.

[35]For the associated mathematical content, see Appendix C.1.

Another factor is platforms' price rigidity on the developer side. One surprising finding from the baseline model is that developers prefer r, the cost savings from multi-homing, to be smaller. This is because, when r increases, platforms proportionally raise the prices they charge developers in order to extract the additional developer surplus. However, in reality, neither iOS nor Android changes its developer-side price as frequently as it changes its consumer-side price.[36] Thus, I extend the baseline model by assuming q_0 and q_1, platforms' developer-side prices, to be exogenously fixed so that each platform's strategy is now a single price posted to consumers. I still use subgame perfect equilibrium as the solution concept and solve for an equilibrium $(\hat{p}_0^*, \hat{p}_1^*, \hat{L}_C^*, \hat{L}_D^*)$ with market shares $(\hat{n}_i^{C*})_{i=0}^1$ and $(\hat{n}_i^{D*})_{i=0}^2$, where $\hat{L}_C^*$ and $\hat{L}_D^*$ still take the form of cutoff strategies specified in (3.3) and (3.4).

The equilibrium prices are

$$\begin{aligned}
\hat{p}_0^* &= \frac{2(st-cd)-mv_+-(d-m)(q_0+q_1)}{2t} - \frac{[st(2c-m)-cd(2c+m)](q_0-q_1)}{2t\hat{\phi}} \\
&\quad + \frac{[st(2c+d-m)-cd(2c+d+m)]v_- + 2t(st-cd-dm)u_-}{2t\hat{\phi}}, \\
\hat{p}_1^* &= \frac{2(st-cd)-mv_+-(d-m)(q_0+q_1)}{2t} + \frac{[st(2c-m)-cd(2c+m)](q_0-q_1)}{2t\hat{\phi}} \\
&\quad - \frac{[st(2c+d-m)-cd(2c+d+m)]v_- + 2t(st-cd-dm)u_-}{2t\hat{\phi}},
\end{aligned} \tag{5.1}$$

where $\hat{\phi} := 3(st-cd)-2dm$.

Combining (5.1) with (A.1.4), the equilibrium market shares are

$$\begin{aligned}
\hat{n}_0^{C*} &= \frac{1}{2} + \frac{(c+m)v_- + tu_- - (c-d+m)(q_0-q_1)}{2\hat{\phi}}, \\
\hat{n}_1^{C*} &= \frac{1}{2} - \frac{(c+m)v_- + tu_- - (c-d+m)(q_0-q_1)}{2\hat{\phi}}, \\
\hat{n}_0^{D*} &= 1 - \frac{v_+-q_0-q_1}{2t} + \frac{(3st-2cd-dm)v_- + tdu_- - (3st-2cd-d^2-dm)(q_0-q_1)}{2t\hat{\phi}}, \\
\hat{n}_1^{D*} &= 1 - \frac{v_+-q_0-q_1}{2t} - \frac{(3st-2cd-dm)v_- + tdu_- - (3st-2cd-d^2-dm)(q_0-q_1)}{2t\hat{\phi}}, \\
\hat{n}_2^{D*} &= \frac{v_+-q_0-q_1}{t} - 1.
\end{aligned} \tag{5.2}$$

The above equilibrium is valid and unique when, as in the baseline model, users are sufficiently horizontally differentiated, the platforms' baseline payoffs to consumers are suf-

[36] As noted in Section 1, on both Android and iOS, the lump-sum fees have remained unchanged for a long time.

ficiently large, and the sum of platforms' baseline payoffs to developers, the magnitudes of users' network effects, the commission, and multi-homers' cost savings are all bounded above and below by different multiples of the degree of developers' horizontal differentiation.[37]

When the above equilibrium is valid and unique, most results stated in Section 4 remain qualitatively valid under this second extension.[38] However, a few results, such as how consumers' switching costs k and multi-homers' cost savings r may affect developer-side prices,[39] are no longer applicable because q_0 and q_1 are exogenously fixed in this extension, and a few others, such as how r may affect platforms' profits and developer surplus,[40] are no longer correct here. Instead, the corresponding new results are as follows.

Proposition 8 (Effects of Multi-homers' Cost Savings on Profits)**.** *When the developer-side prices are fixed, an increase in multi-homers' cost savings r increases (decreases) a platform's equilibrium profit if and only if the platform charges developers a positive (negative) price.*

Proposition 8 indicates that when the developer-side prices are fixed, since platforms can no longer adjust their respective developer-side prices in response to any change in multi-homers' cost savings r, the relevant effect on each platform's profit is purely extensive. In particular, the effect of an increase in r on each platform is only via the additional developers that choose to operate on it after the increase.[41] If a platform charges a positive developer-side price, the platform would benefit from an increase in r, which increases the number of developers operating on it and paying it. However, when a platform subsidizes the developers operating on it—that is, it charges a negative developer-side price—the net effect of an increase in r is a larger total subsidy that the platform provides to the developers operating on it, thus being negative. Similarly, a zero developer-side price of a platform implies the platform's indifference to any change in r. The other two parts of r's effect, which are respectively via the consumer-side price and the commission, exactly offset each other.[42]

[37] For the steps for solving this extended model and the associated mathematical content, including the expressions associated with these conditions, see Appendix C.2.

[38] However, some of them exhibit minor quantitative differences. The proofs of all valid propositions, the details about the quantitative differences, and the proofs of Propositions 8 and 9 are relegated to Appendix C.3.

[39] These results correspond to part 2 of Proposition 1 and part 2 of Proposition 2.

[40] These results correspond to Proposition 4 and part 2 of Proposition 7. They may also affect the validity of part 2 of Corollary 2 and of Corollary 3.

[41] The fraction of developers that operate on platform $i \in \{0,1\}$ is $\hat{n}_i^{D*} + \hat{n}_2^{D*}$, where $\frac{\partial}{\partial r}(\hat{n}_i^{D*} + \hat{n}_2^{D*}) = -\frac{1}{t} + \frac{2}{t} = \frac{1}{t}$, and $\frac{\partial \hat{\pi}_i^*}{\partial r} = \frac{q_i}{t}$, where $\hat{\pi}_i^*$ is the equilibrium profit of platform i.

[42] The part of the profit that comes from the consumer-side price is $\hat{p}_i^* \hat{n}_i^{C*}$, where $\frac{\partial(\hat{p}_i \hat{n}_i^{C*})}{\partial r} = -\frac{m\hat{n}_i^{C*}}{t}$, and the part that comes from the commission is $m\hat{n}_i^{C*}(\hat{n}_i^{D*} + \hat{n}_2^{D*})$, where $\frac{\partial[m\hat{n}_i^{C*}(\hat{n}_i^{D*}+\hat{n}_2^{D*})]}{\partial r} = \frac{m\hat{n}_i^{C*}}{t} = -\frac{\partial(\hat{p}_i \hat{n}_i^{C*})}{\partial r}$.

Proposition 9 (Effects of Multi-homers' Cost Savings on Developer Surplus)**.** *When the developer-side prices are fixed, an increase in multi-homers' cost savings r increases the equilibrium developer surplus.*

Proposition 9 shows that when the developer-side prices are fixed, the platforms' inability to adjust the prices they charge developers in response to changes in multi-homers' cost savings r reverses the direction of r's effect on developers. In particular, the effect of an increase in r is now simply that every multi-homer fully benefits from the increase.[43] Single-homers' payoffs are invariant to any change in r.

Corollary 4 (Effects of Restrictions on Super Apps on Developer Surplus)**.** *When the developer-side prices are fixed, in response to restrictions on super apps, the equilibrium developer surplus*

1. *decreases when consumers are sufficiently horizontally differentiated;*

2. *may increase when consumers are not very horizontally differentiated and the restrictions increase consumers' switching costs to a significantly greater extent than they decrease multi-homers' cost savings.*

Corollary 4 is about the effect of restrictions on super apps on the equilibrium developer surplus. This corollary follows directly from Proposition 9 and Proposition 6, and each part of it is similar to the situation in the corresponding part of Corollary 3. Part 1 shows how the effect is like when consumers are sufficiently horizontally differentiated, the effect on the equilibrium developer surplus operating through the decrease in multi-homers' cost savings outweighs the effect operating through the increase in consumers' switching costs—the latter effect is relatively small. However, Proposition 9 shows that the former effect is negative when the developer-side prices are fixed. Therefore, the overall effect of such restrictions on the equilibrium developer surplus is negative as well.

Part 2 qualitatively describes the ambiguity associated with the effect of the restrictions on developers when consumers are not very horizontally differentiated. Unlike in Corollary 3, under this extension, the equilibrium developer surplus increases in both consumers' switching costs and multi-homers' cost savings when the strong platform serves more developers in equilibrium than the weak platform. In this situation, when the restrictions increase consumers' switching costs to a significantly greater extent than they decrease multi-homers' cost savings, the positive effect of the restrictions operating through consumers' switching

[43] Similar to the situation in Proposition 7, this explanation is also consistent with the corresponding quantitative results: the fraction of multi-homers is $\hat{n}_2^{D*}$, and each of them would fully enjoy any increase in r, corresponding to $\frac{\partial \check{DS}^*}{\partial r} = \hat{n}_2^{D*} > 0$.

costs may outweigh the negative effect operating through multi-homers' cost savings. As a result, the overall effect of the restrictions on the equilibrium developer surplus may be positive.

In conclusion, this second extension accounts for platforms' developer-side price rigidity in reality by holding their developer-side prices exogenously fixed. Such price rigidity disables platforms from extracting surplus by charging higher prices to developers when there is any increase in multi-homers' cost savings. Because platforms' developer-side prices are almost always non-negative in practice—that is, platforms hardly ever subsidize developers—a platform tends to be better off when multi-homers' cost savings increase. Therefore, combined with part 1 of Proposition 3 and similar to the baseline model, a platform strong on the consumer side may find restrictions on super apps beneficial, whereas a platform that is not strong does not prefer any such restrictions.[44] Overall, this result is qualitatively consistent with Corollary 2, the corresponding finding in the baseline model. Quantitatively, however, a strong platform may find these restrictions more attractive than in the baseline model, since it loses less via the channel of its developer-side price from the resulting reduction in multi-homers' cost savings. This is because developer-side price rigidity limits platforms' ability to reduce developer-side prices in response to any reduction in multi-homers' cost savings. The above new results under this extension further corroborate the possible profitability of the restrictions on super apps for a platform holding a dominant position on the consumer side, like Apple's iOS in the United States, and they do not fundamentally affect the validity of the rival-related aspect of Apple's potential antitrust concerns in the United States.

In contrast, under this extension, regardless of the values of the developer-side prices, developers' attitude toward multi-homers' cost savings qualitatively differs from that in the baseline model: the equilibrium developer surplus increases in multi-homers' cost savings. Therefore, as indicated in part 1 of Corollary 4, when perceive the platforms as highly differentiated, developers become worse off under restrictions on super apps. This conclusion points to an additional aspect of Apple's potential antitrust concerns in the United States: Apple's restrictions on super apps may also adversely affect developers.

On the other hand, developers may also benefit from these restrictions. In particular, the direction of these restrictions' effect on developers depends on the contrast in the extent to which the restrictions on super apps affect consumers' switching costs and multi-homers'

[44] Quantitatively, as an exception, under this extension, when neither platform is strong and the developer-side price of a platform i is zero (i.e., $f = 0.5$ and $q_i = 0$), this platform is indifferent to restrictions on super apps. This is because the effect of the restrictions via the channel of consumers' switching costs is $\frac{\partial \hat{\pi}_i^*}{\partial k} = \frac{(-1)^{1[i=1]}(2f-1)[2(st-cd)-dm]\hat{n}_i^{C*}}{\hat{\phi}} = 0$, and that operating through multi-homers' cost savings is $\frac{\partial \hat{\pi}_i^*}{\partial r} = \frac{q_i}{t} = 0$.

cost savings—which Section 4 also identifies as determining the direction of the effect on consumers—and on the contrast between the platforms' equilibrium developer-side market shares. From Corollary 2, Apple has an incentive to restrict super apps only when the restrictions increase consumers' switching costs to a significantly greater extent than they decrease multi-homers' cost savings. In this situation, when iOS serves sufficiently more developers than its rivals, developer surplus may decrease in consumers' switching costs. Therefore, developers may benefit from Apple's restrictions on super apps when the effect operating through consumers' switching costs outweighs that operating through multi-homers' cost savings.

6 Concluding Remarks

This paper investigates a two-sided market where platforms compete for two groups of users—consumers and developers—that enjoy cross-side network effects. Each consumer single-homes, begins as a patron of one platform, and incurs a switching cost when choosing to switch to the other platform. Developers may choose to multi-home, and each multi-homer enjoys a non-negative cost saving. The two groups of users' baseline payoffs from choosing each platform, the degrees of horizontal differentiation, the magnitudes of network effects, consumers' switching costs, multi-homers' cost savings, platforms' consumer-side initial market shares, and the commission each platform charges are all exogenous. These parameters are key objects to be estimated in the relevant empirical analysis. I put this framework in the context of Apple's antitrust concerns in the smartphone platform market, characterize the market equilibrium, and examine the effects of super apps and the relevant restrictions on the equilibrium. The paper's findings explain why Apple's attitude toward super apps varies across regions and reveal three possible distinct dimensions of Apple's potential antitrust concerns in the United States, which pertain respectively to iOS's rivals, consumers, and developers.

Unlike previous studies that usually consider Apple's full control over the iOS App Store to be a market power that mainly affects the developer side of the smartphone platform market, I identify super apps as a channel via which this market power can directly affect the consumer side as well. I also find that Apple's U.S. restrictions on super apps may indeed give rise to antitrust concerns. My model considers these restrictions a hybrid of technological tying and refusals to deal. They benefit Apple, whose iOS platform holds a dominant position on the consumer side of the U.S. smartphone platform market, while harming iOS's rival platforms in the United States, which are comparatively weak on the consumer side. In contrast, Apple does not have a strong incentive to intensively restrict

super apps in Asia, where iOS is weak on the consumer side. When consumers perceive the platforms as highly differentiated, these restrictions adversely affect consumers in the United States via two channels: directly via consumers' switching costs and indirectly via multi-homers' cost savings. When I extend the model to incorporate platforms' developer-side price rigidity, such restrictions also adversely affect developers because they reduce multi-homers' cost savings.

When consumers perceive the platforms as highly differentiated, they may, on the other hand, be better off under the restrictions—exactly in the circumstances that make restricting super apps profitable for Apple in the United States: when the restrictions increase their switching costs to a significantly greater extent than they decrease multi-homers' cost savings. In this situation, when there are more apps on Apple's iOS—which is strong in the United States—than on its rival platforms, developers may be better off under such restrictions as well. Evaluating these dimensions of the potential antitrust concerns associated with Apple's restrictions on super apps in the United States therefore requires closer empirical examination of these relevant objects.

The existing literature on two-sided markets that incorporate both switching costs and other influencing factors remains relatively sparse. This paper is, to the best of my knowledge, the first to analytically examine a setting in which switching costs and another factor—multi-homers' cost savings—simultaneously shape the equilibrium in a two-sided market. However, substantial work remains in this field. First, my model focuses on a static setting, whereas the smartphone platform market is dynamic in reality. Although the relatively short history of the current smartphone platform market justifies the usage of initial market shares in my model, the restrictions on super apps and their associated effects tend to persist for long periods of time. Even if the effects found in a dynamic setting do not qualitatively differ from those in my static model, a quantitative comparison between short-term and long-term effects may still yield valuable insights. Second, iOS and Android compete not only in the smartphone platform market but also in the tablet market, and Apple's restrictions on super apps simultaneously affect both markets. When modeling competition in the tablet market together with that in the smartphone platform market, the situation may become more intricate because consumers' platform choices may vary across devices and their switching costs could depend on these choices. Additional complexity would arise from the fact that tablet ownership is incomplete among consumers active in the smartphone platform market. The above considerations point to promising directions for extending models to analyze the joint effects of switching costs and other factors in two-sided markets, whether in the smartphone industry or in other sectors.

A Mathematical Content Omitted in Section 3

A.1 Steps for Solving the Baseline Model

To find an equilibrium $[(p_0^*, q_0^*), (p_1^*, q_1^*), L_C^*, L_D^*]$ with market shares $(n_i^{C*})_{i=0}^1$ and $(n_i^{D*})_{i=0}^2$, I solve the model from the second stage of the game.

First, I solve for $(n_i^{C*})_{i=0}^1$ and $(n_i^{D*})_{i=0}^2$ from the interaction between users' optimal platform choices. Because platforms' prices $(p_i, q_i)_{i=0}^1$ are given at this stage, let x_0^* and x_1^* denote the cutoff values of consumers so that the equilibrium strategy of a consumer with $(x, b) \in [0,1] \times \{0,1\}$ is

$$L_C^*(x, b, p_0, q_0, p_1, q_1) = \begin{cases} 0 & \text{if } x \le x_b^* \\ 1 & \text{elsewhere} \end{cases}, \tag{3.3}$$

which requires $U_0(x_b^*, b) = U_1(x_b^*, b)$ for each $b \in \{0,1\}$. Combining this condition with (2.1) yields

$$\begin{aligned} x_0^* &= \frac{1}{2} + \frac{u_0 - u_1 - p_0 + p_1 + c(n_0^{D*} - n_1^{D*}) + k}{2s}, \\ x_1^* &= \frac{1}{2} + \frac{u_0 - u_1 - p_0 + p_1 + c(n_0^{D*} - n_1^{D*}) - k}{2s}. \end{aligned} \tag{A.1.1}$$

Because consumers' horizontal preferences are independent of the platforms they initially patronize, the resulting consumer-side market shares are

$$\begin{aligned} n_0^{C*} &= f x_0^* + (1-f) x_1^* = \frac{1}{2} + \frac{u_0 - u_1 - p_0 + p_1 + c(n_0^{D*} - n_1^{D*}) + k(2f-1)}{2s}, \\ n_1^{C*} &= 1 - n_0^{C*} = \frac{1}{2} - \frac{u_0 - u_1 - p_0 + p_1 + c(n_0^{D*} - n_1^{D*}) + k(2f-1)}{2s}. \end{aligned} \tag{A.1.2}$$

Note that the expressions of $(n_i^{C*})_{i=0}^1$ in (A.1.2) are well-defined even though they contain $(n_i^{D*})_{i=0}^1$, which are obtained from developers' optimal platform choices. This is because in equilibrium consumers take developers' strategies as given when optimizing their own choices.

Similarly, let n_0^{D*} and $1 - n_1^{D*}$ denote the cutoff values of developers so that the equilibrium strategy of a developer with $y \in [0,1]$ is

$$L_D^*(y, p_0, q_0, p_1, q_1) = \begin{cases} 0 & \text{if } y < n_0^{D*} \\ 1 & \text{if } y > 1 - n_1^{D*} \\ 2 & \text{elsewhere} \end{cases}, \tag{3.4}$$

which requires $V_0(n_0^{D*}) = V_2(n_0^{D*})$ and $V_1(1-n_1^{D*}) = V_2(1-n_1^{D*})$. Combining this condition with (2.2) and (2.3) yields

$$\begin{aligned} n_0^{D*} &= 1 - \frac{v_1 - q_1 + dn_1^{C*} + r}{t}, \\ n_1^{D*} &= 1 - \frac{v_0 - q_0 + dn_0^{C*} + r}{t}, \\ n_2^{D*} &= 1 - n_0^{D*} - n_1^{D*}, \end{aligned} \tag{A.1.3}$$

which are also well-defined by the same reasoning as for (A.1.2).

Combining (A.1.2) and (A.1.3) yields a system of five equations for five unknowns. Solving this equation system yields

$$\begin{aligned} n_0^{C*} &= \frac{1}{2} + \frac{c(v_- - q_0 + q_1) + t(u_- - p_0 + p_1)}{2(st - cd)}, \\ n_1^{C*} &= \frac{1}{2} - \frac{c(v_- - q_0 + q_1) + t(u_- - p_0 + p_1)}{2(st - cd)}, \\ n_0^{D*} &= 1 - \frac{v_+ - q_0 - q_1}{2t} + \frac{s(v_- - q_0 + q_1) + d(u_- - p_0 + p_1)}{2(st - cd)}, \\ n_1^{D*} &= 1 - \frac{v_+ - q_0 - q_1}{2t} - \frac{s(v_- - q_0 + q_1) + d(u_- - p_0 + p_1)}{2(st - cd)}, \\ n_2^{D*} &= \frac{v_+ - q_0 - q_1}{t} - 1, \end{aligned} \tag{A.1.4}$$

where $u_- := u_0 - u_1 + k(2f-1)$, $v_- := v_0 - v_1$, and $v_+ := v_0 + v_1 + d + 2r$.

Next, I solve for $(p_i^*, q_i^*)_{i=0}^1$ from platforms' profit maximization problems. For each platform $i \in \{0,1\}$, the problem is

$$\max_{p_i, q_i} \pi_i = p_i n_i^{C*} + (q_i + mn_i^{C*})(n_i^{D*} + n_2^{D*}). \tag{A.1.5}$$

The first-order conditions are

$$\begin{aligned} \frac{\partial \pi_i}{\partial p_i} &= n_i^{C*} - \frac{t[p_i + m(n_i^{D*} + n_2^{D*})]}{2(st - cd)} - \frac{d(q_i + mn_i^{C*})}{2(st - cd)} = 0, \\ \frac{\partial \pi_i}{\partial q_i} &= n_i^{D*} + n_2^{D*} - \frac{c[p_i + m(n_i^{D*} + n_2^{D*})]}{2(st - cd)} - (q_i + mn_i^{C*})\Big[\frac{s}{2(st - cd)} + \frac{1}{2t}\Big] = 0. \end{aligned} \tag{A.1.6}$$

The second-order conditions are

$$\frac{\partial^2 \pi_i}{\partial p_i^2} < 0, \ \frac{\partial^2 \pi_i}{\partial q_i^2} < 0, \text{ and } \frac{\partial^2 \pi_i}{\partial p_i^2}\frac{\partial^2 \pi_i}{\partial q_i^2} - \Big(\frac{\partial^2 \pi_i}{\partial p_i q_i}\Big)^2 > 0, \tag{A.1.7}$$

where

$$\begin{aligned}\frac{\partial^2\pi_i}{\partial p_i^2} &= -\frac{t[2(st-cd)-dm]}{2(st-cd)^2},\\ \frac{\partial^2\pi_i}{\partial q_i^2} &= -\frac{(2st-cd)[2(st-cd)-cm]}{2t(st-cd)^2},\\ \frac{\partial^2\pi_i}{\partial p_i q_i} &= -\frac{c+d-\frac{m}{2}}{2(st-cd)}+\frac{m(st+cd)}{4(st-cd)^2}.\end{aligned} \tag{A.1.8}$$

From (A.1.7) and (A.1.8), the second-order conditions hold whenever $st > d(c+\frac{m}{2})$, $st > c(d+\frac{m}{2})$ and $8st-4cd-(m+c+d)^2 > 0$. A sufficient condition for these inequalities is

$$\min\{s,t\} > \max\{c,d\} + \frac{m}{2}, \tag{A.1.9}$$

which I assume throughout this paper. Both this inequality and the above conditions imply that the degrees of users' horizontal differentiation are relatively large compared with the magnitudes of users' network effects.

The unique solution to the four-equation system constituted by (A.1.6) is $(p_0^*, q_0^*, p_1^*, q_1^*)$,[45] whose closed-form expressions are in (3.1). (A.1.9) guarantees that this solution indeed maximizes each platform's profit.

When the above solution constitutes an equilibrium, the equilibrium would be characterized by the prices in (3.1), the market shares in (3.2), and users' strategies in (3.3) and (3.4), where, from (A.1.1) and (A.1.2), $x_0^* = n_0^{C*} + \frac{k(1-f)}{s}$, $x_1^* = n_0^{C*} - \frac{kf}{s}$.

A.2 Conditions for Validity and Uniqueness of the Equilibrium

(3.1) presents the unique solution to the equation system given by (A.1.6), and this solution constitutes a unique equilibrium of the game when it is valid. Because the solution is obtained from the first-order conditions, it is valid if it is indeed an interior solution. This requires the following conditions in addition to (A.1.9):

1. Consumers' participation constraint.

   In equilibrium, every consumer prefers using a platform to not using any platform. Equivalently, $U_0(x_b^*, b) = U_1(x_b^*, b) \geq 0$ for each $b \in \{0,1\}$, which holds under

$$u_0 + u_1 - k + \frac{(c+d+m)v_+ - 6st + c^2 + 3cd + m(2c+d) + m^2}{2t} \geq 0. \tag{A.2.1}$$

[45] (A.1.6) consists of two equations for each $i \in \{0,1\}$, giving a system of four equations.

2. Developers' participation constraint.

   In equilibrium, every developer prefers operating on at least one platform to operating on neither.[46] Equivalently, $V_0(n_0^{D*}) = V_2(n_0^{D*}) \geq 0$ and $V_1(1 - n_1^{D*}) = V_2(1 - n_1^{D*}) \geq 0$, which hold under

$$v_+ + c + m - 2r - 2t \geq 0. \tag{A.2.2}$$

3. Interior consumer-side market shares.

   The equilibrium satisfies $x_0^*, x_1^* \in (0,1)$, meaning that each platform serves some consumers that initially patronize itself and serves some consumers that initially patronize its rival. This implies $n_0^{C*}, n_1^{C*} \in (0,1)$ and holds under

$$-1 + \frac{2kf}{s} < \left|\frac{(c+d+m)v_- + 2tu_-}{\phi}\right| < 1 - \frac{2k(1-f)}{s}. \tag{A.2.3}$$

4. Interior developer-side market shares.

   The equilibrium satisfies $n_0^{D*}, n_1^{D*} \in (0,1)$, meaning that each platform serves some single-homing developers.[47] Given (A.2.2), which implies $\frac{4t-(v_++c+m)}{2} < \frac{v_++c+m}{2}$, this holds under

$$\left|\frac{(3st - cd)v_- + t(c+d+m)u_-}{\phi}\right| < \frac{4t - (v_+ + c + m)}{2}. \tag{A.2.4}$$

In summary, the above conditions require that

1. the degrees of users' horizontal differentiation are relatively large compared with the differences in platforms' baseline payoffs on both sides of the market and consumers' switching costs,

2. the platforms' baseline payoffs to consumers are relatively large compared with consumers' switching costs and the degree of consumers' horizontal differentiation, and

3. the sum of platforms' baseline payoffs to developers, the magnitudes of users' network effects, the commission, and multi-homers' cost savings are all bounded above and below by different multiples of the degree of developers' horizontal differentiation.

[46] Because $r \geq 0$, this condition also guarantees $n_2^{D*} = \frac{v_++c+m-2t}{2t} \geq 0$, i.e., the number of multi-homing developers is non-negative.

[47] This condition implicitly requires $v_+ + c + m < 4t$, which is equivalent to $n_2^{D*} = \frac{v_++c+m-2t}{2t} < 1$, i.e., not every developer multi-homes.

These conditions guarantee the validity of the interior solution shown in (3.1). In the context of the model, they ensure that the equilibrium is non-trivial: neither platform serves the entire population of users on either side of the market. The equilibria for the extended models in Section 5 are also valid under conditions of the same type, and those equilibria are likewise non-trivial.

B Proofs Omitted in Section 4

Unless specified otherwise, in all proofs presented in Appendices B and C, I assume without loss of generality that $f \geq 0.5$. This means that, on the consumer side of the market, either platform 0 holds a dominant position or neither platform is strong.

B.1 Proof of Proposition 1

B.1.1 Part 1

From (3.1), with $u_- = u_0 - u_1 + k(2f-1)$,

$$\frac{\partial p_0^*}{\partial k} = -\frac{\partial p_1^*}{\partial k} = \frac{[4st - d(3c+d) - m(c+2d) - m^2](2f-1)}{2\phi}. \tag{B.1.1}$$

Under (A.1.9),

$$\begin{aligned} 4st - d(3c+d) - m(c+2d) - m^2 &> 3(c+\frac{m}{2})(d+\frac{m}{2}) + (d+\frac{m}{2})^2 - d(3c+d) - m(c+2d) - m^2 \\ &= m(\frac{3}{2}d + \frac{3}{2}c + d) + m^2 - m(c+2d) - m^2 \\ &= \frac{(c+d)m}{2} \geq 0, \end{aligned} \tag{B.1.2}$$

and

$$\begin{aligned} \phi &= 6st - c^2 - d^2 - 4cd - 2m(c+d) - m^2 \\ &> (c+\frac{m}{2})^2 + 4(c+\frac{m}{2})(d+\frac{m}{2}) + (d+\frac{m}{2})^2 - c^2 - d^2 - 4cd - 2m(c+d) - m^2 \\ &= m(c+2c+2d+d) + \frac{3}{2}m^2 - 2m(c+d) - m^2 \\ &= (c+d)m + \frac{m^2}{2} \geq 0. \end{aligned} \tag{B.1.3}$$

Therefore, combining (B.1.1)–(B.1.3) yields $\frac{\partial p_0^*}{\partial k} = -\frac{\partial p_1^*}{\partial k} \geq 0$, where the weak inequality

binds if and only if $f = 0.5$. Accordingly, when k increases, a platform that is strong (weak) on the consumer side charges a higher (lower) price to consumers. □

B.1.2 Part 2

From (3.1), with $u_- = u_0 - u_1 + k(2f-1)$,

$$\frac{\partial q_0^*}{\partial k} = -\frac{\partial q_1^*}{\partial k} = \frac{t(d-c-m)(2f-1)}{2\phi}. \tag{4.1.1}$$

Accordingly, when k increases, a platform that is strong (weak) on the consumer side charges a higher (lower) price to developers if and only if $d > c+m$—that is, the magnitude of the developer-side network effect d is sufficiently large relative to that of the consumer-side network effect c and the commission m. □

B.1.3 Part 3

From (3.2), with $u_- = u_0 - u_1 + k(2f-1)$,

$$\frac{\partial n_0^{C*}}{\partial k} = -\frac{\partial n_1^{C*}}{\partial k} = \frac{t(2f-1)}{\phi} \geq 0 \text{ by (B.1.3)}, \tag{B.1.4}$$

where the weak inequality binds if and only if $f = 0.5$. Accordingly, when k increases, a platform that is strong (weak) on the consumer side serves more (fewer) consumers. □

B.1.4 Part 4

From (3.2), $\frac{\partial n_2^{D*}}{\partial k} = 0$. Also, combining (3.2) with $u_- = u_0 - u_1 + k(2f-1)$ yields

$$\frac{\partial n_0^{D*}}{\partial k} = -\frac{\partial n_1^{D*}}{\partial k} = \frac{(c+d+m)(2f-1)}{2\phi} \geq 0 \text{ by (B.1.3)}, \tag{B.1.5}$$

where the weak inequality binds if and only if $f = 0.5$. Accordingly, when k increases, a platform that is strong (weak) on the consumer side serves more (fewer) developers, and the number of multi-homers remains unchanged. □

B.2 Proof of Proposition 2

B.2.1 Part 1

From (3.1), with $v_+ = v_0 + v_1 + d + 2r$,

$$\frac{\partial p_0^*}{\partial r} = \frac{\partial p_1^*}{\partial r} = \frac{-2(d+m)}{4t} = -\frac{d+m}{2t} < 0. \tag{B.2.1}$$

Accordingly, when r increases, each platform charges a lower price to consumers. □

B.2.2 Part 2

From (3.1), with $v_+ = v_0 + v_1 + d + 2r$,

$$\frac{\partial q_0^*}{\partial r} = \frac{\partial q_1^*}{\partial r} = \frac{2}{4} = \frac{1}{2} > 0. \tag{B.2.2}$$

Accordingly, when r increases, each platform charges a higher price to developers. □

B.2.3 Part 3

From (3.2),

$$\frac{\partial n_0^{C*}}{\partial r} = \frac{\partial n_1^{C*}}{\partial r} = 0. \tag{B.2.3}$$

Accordingly, when r increases, each platform still serves the same number of consumers. □

B.2.4 Part 4

From (3.2), with $v_+ = v_0 + v_1 + d + 2r$,

$$\begin{aligned} &\frac{\partial n_0^{D*}}{\partial r} = \frac{\partial n_1^{D*}}{\partial r} = -\frac{2}{4t} = -\frac{1}{2t} < 0, \\ &\frac{\partial n_2^{D*}}{\partial r} = \frac{2}{2t} = \frac{1}{t} = \left|\frac{\partial n_0^{D*}}{\partial r}\right| + \left|\frac{\partial n_1^{D*}}{\partial r}\right| > 0. \end{aligned} \tag{B.2.4}$$

Accordingly, when r increases, each platform serves more developers, and on each platform, some developers switch from single-homing to multi-homing. □

B.3 Proof of Proposition 3

From (4.1.1), (4.2.1), (B.1.1), (B.1.4), and (B.1.5),

$$\begin{aligned}\frac{\partial \pi_0^*}{\partial k} &= \frac{2f-1}{2\phi}\{[4st - d(3c+d) - m(c+2d) - m^2]n_0^{C*} \\ &\qquad + 2tp_0^* + (c+d+m)(q_0^* + mn_0^{C*}) + t(d-c-m+2m)(n_0^{D*} + n_2^{D*})\} \qquad \text{(B.3.1)} \\ &= \frac{(2f-1)[8st - c^2 - d^2 - 6cd - 2m(c+d) - m^2]n_0^{C*}}{2\phi}.\end{aligned}$$

Because (1) (A.2.3) implies that $n_0^{C*} > 0$ and (2) combining (B.1.3) and (A.1.9) implies that $8st - c^2 - d^2 - 6cd - 2m(c+d) - m^2 > 2(st - cd) > 0$, it follows that $\frac{\partial \pi_0^*}{\partial k} \geq 0$, where the weak inequality binds if and only if $f = 0.5$.

Symmetrically,

$$\begin{aligned}\frac{\partial \pi_1^*}{\partial k} &= \frac{-(2f-1)}{2\phi}\{[4st - d(3c+d) - m(c+2d) - m^2]n_1^{C*} \\ &\qquad + 2tp_1^* + (c+d+m)(q_1^* + mn_1^{C*}) + t(d-c-m+2m)(n_1^{D*} + n_2^{D*})\} \qquad \text{(B.3.2)} \\ &= \frac{-(2f-1)[8st - c^2 - d^2 - 6cd - 2m(c+d) - m^2]n_1^{C*}}{2\phi} \leq 0,\end{aligned}$$

and the weak inequality binds if and only if $f = 0.5$.

Accordingly, when k increases, the equilibrium profit of a platform increases (decreases) if and only if the platform is strong (weak) on the consumer side. □

B.4 Proof of Proposition 4

From (4.2.1) and (B.2.1)–(B.2.4),

$$\begin{aligned}\frac{\partial \pi_0^*}{\partial r} &= \frac{n_0^{D*} + n_2^{D*}}{2} - \frac{(d+m)n_0^{C*}}{2t} + \frac{p_0 + mn_0^{C*}}{2t} = \frac{v_0 + r}{2t} > 0, \\ \frac{\partial \pi_1^*}{\partial r} &= \frac{n_1^{D*} + n_2^{D*}}{2} - \frac{(d+m)n_1^{C*}}{2t} + \frac{p_1 + mn_1^{C*}}{2t} = \frac{v_1 + r}{2t} > 0.\end{aligned} \qquad \text{(B.4.1)}$$

Accordingly, the equilibrium profit of each platform increases in r. □

B.5 Proof of Corollary 2

Consider restrictions on super apps that raise consumers' switching costs by $\Delta k > 0$—from some $k_0 \geq 0$ to $k_0 + \Delta k$—and reduce multi-homers' cost savings by $\Delta r > 0$—from some $r_0 > 0$ to $r_0 - \Delta r \geq 0$—such that (A.2.1)–(A.2.4) hold for all $k \in [k_0, k_0 + \Delta k]$ and all $r \in [r_0 - \Delta r, r_0]$.

Also, from (3.2), (B.2.3), (B.3.1), and (B.4.1), for each $i \in \{0,1\}$, the value of $\frac{\partial \pi_i^*}{\partial k}$ is independent of r, and the value of $\frac{\partial \pi_i^*}{\partial r}$ is independent of k. Denote the overall effect of the restrictions on super apps on platform i's equilibrium profit by

$$\Delta\pi_i^* = \Delta k \int_{k_0}^{k_0+\Delta k} \frac{\partial \pi_i^*}{\partial k} dk - \Delta r \int_{r_0}^{r_0-\Delta r} \frac{\partial \pi_i^*}{\partial r} dr. \tag{B.5.1}$$

B.5.1 Part 1

Because $f \geq 0.5$, platform 1 is not strong on the consumer side.

Then, from (3.2), (B.2.3), (B.3.1), and (B.4.1), for any $(k,r) \in [k_0, k_0+\Delta k] \times [r_0-\Delta r, r_0]$,

$$\begin{aligned} &\frac{\partial \pi_1^*}{\partial r} \in [\frac{v_1 + r_0 - \Delta r}{2t}, \frac{v_1 + r_0}{2t}] \subseteq (0, +\infty), \\ &\frac{\partial \pi_1^*}{\partial k} \in [\frac{\partial \pi_1^*}{\partial k}\Big|_{k=k_0}, \frac{\partial \pi_1^*}{\partial k}\Big|_{k=k_0+\Delta k}] \subseteq (-\infty, 0]. \end{aligned} \tag{B.5.2}$$

Therefore, from (B.5.1) and (B.5.2),

$$\begin{aligned} \Delta\pi_1^* &= \Delta k \int_{k_0}^{k_0+\Delta k} \frac{\partial \pi_1^*}{\partial k} dk - \Delta r \int_{r_0}^{r_0-\Delta r} \frac{\partial \pi_1^*}{\partial r} dr \\ &\leq -\Delta r \int_{r_0}^{r_0-\Delta r} \frac{\partial \pi_i^*}{\partial r} dr \text{ (because } \frac{\partial \pi_1^*}{\partial k} \leq 0 \text{ for all } k \in [k_0, k_0+\Delta k]) \\ &< 0 \text{ (because } \frac{\partial \pi_1^*}{\partial r} > 0 \text{ for all } r \in [r_0 - \Delta r, r_0]). \end{aligned} \tag{B.5.3}$$

Thus, the equilibrium profit of a platform that is not strong on the consumer side decreases in response to restrictions on super apps. □

B.5.2 Part 2

First, still without loss of generality but for this proof only, assume $f > 0.5$, so that platform 0 is (strictly) strong on the consumer side.

Second, from (3.2), (B.2.3), (B.3.1), and (B.4.1), for any $(k,r) \in [k_0, k_0+\Delta k] \times [r_0-\Delta r, r_0]$,

$$\begin{aligned} &\frac{\partial \pi_0^*}{\partial r} \in [\frac{v_0 + r_0 - \Delta r}{2t}, \frac{v_0 + r_0}{2t}] \subseteq (0, +\infty), \\ &\frac{\partial \pi_0^*}{\partial k} \in [\frac{\partial \pi_0^*}{\partial k}\Big|_{k=k_0}, \frac{\partial \pi_0^*}{\partial k}\Big|_{k=k_0+\Delta k}] \subseteq (0, +\infty). \end{aligned} \tag{B.5.4}$$

Thus, from (B.5.1) and (B.5.4),

$$\begin{aligned}\Delta\pi_0^* &\geq \Delta k \frac{\partial \pi_0^*}{\partial k}\Big|_{k=k_0} - \Delta r \frac{v_0 + r_0}{2t} \\ &> 0 \text{ if } \Delta k > \frac{\Delta r \frac{v_0+r_0}{2t}}{\frac{\partial \pi_0^*}{\partial k}\Big|_{k=k_0}}.\end{aligned} \tag{B.5.5}$$

Similarly,

$$\begin{aligned}\Delta\pi_0^* &\leq \Delta k \frac{\partial \pi_0^*}{\partial k}\Big|_{k=k_0+\Delta k} - \Delta r \frac{v_0 + r_0 - \Delta r}{2t} \\ &< 0 \text{ if } \Delta k < \frac{\Delta r \frac{v_0+r_0-\Delta r}{2t}}{\frac{\partial \pi_0^*}{\partial k}\Big|_{k=k_0+\Delta k}}.\end{aligned} \tag{B.5.6}$$

From (B.5.5), $\Delta\pi_0^* > 0$ whenever Δk is sufficiently large relative to Δr. Similarly, from (B.5.6), $\Delta\pi_0^* < 0$ whenever Δk is sufficiently small relative to Δr. Accordingly, the profit of a strong platform increases (decreases) in response to restrictions on super apps when the restrictions increase consumers' switching costs to a significantly greater (lesser) extent than they decrease multi-homers' cost savings. □

B.6 Proof of Part 1 of Proposition 5

First, part 2 of Proposition 5 entails ambiguity and thus does not come with a proof. The relevant simulation results are in Figure 2. Below is the proof for part 1.

From (4.2.2), (B.1.1), (B.1.4), and (B.1.5), and because (A.2.3) gives $n_0^{C*} < 1$, for any $f \in [0.5, 1)$,

$$\begin{aligned}\frac{\partial CS^*}{\partial k} &= \frac{[-c(c+d+m) + 4st - d(3c+d) - m(c+2d) - m^2](2f-1)}{2\phi} \\ &\quad + \frac{2stn_0^{C*}(2f-1)}{\phi} - f + \frac{2kf(1-f)}{s} \\ &\leq \frac{[4st - c^2 - d^2 - 4cd - 2m(c+d) - m^2](2f-1)}{2\phi} + \frac{2st(2f-1)}{\phi} - f + \frac{2kf(1-f)}{s} \\ &= -\frac{1}{2} + \frac{st(2f-1)}{\phi} + \frac{2kf(1-f)}{s}.\end{aligned} \tag{B.6.1}$$

Define $g(f) := -\frac{1}{2} + \frac{st(2f-1)}{\phi} + \frac{2kf(1-f)}{s}$ for all $f \in [0.5, 1)$, then

$$\begin{aligned} g'(f) &= \frac{2st}{\phi} + \frac{2k(1-2f)}{s}, \\ g''(f) &= -\frac{4k}{s} \leq 0. \end{aligned} \tag{B.6.2}$$

From (B.1.3), $g'(0.5) = \frac{2st}{\phi} > 0$. Thus, to determine the maximum value of $g(f)$ on $[0.5, 1)$, I consider the following two scenarios.

Scenario 1: $g'(1) \geq 0$

Since $g''(f) = -\frac{4k}{s} \leq 0$, $g'(f) \geq 0$ for all $f \in [0.5, 1)$. Therefore,

$$\max_{f \in [0.5,1)} g(f) = g(1) = -\frac{1}{2} + \frac{st}{\phi} < 0 \text{ if and only if } 2st < \phi, \tag{B.6.3}$$

i.e., $s > \frac{c^2+d^2+4cd+2m(c+d)+m^2}{4t}$.

Scenario 2: $g'(1) < 0$

In this scenario,

$$0 > g'(1) = \frac{2st}{\phi} - \frac{2k}{s}, \tag{B.6.4}$$

i.e., $\frac{k}{s} > \frac{st}{\phi}$.

Since $g'(0.5) > 0$ and $g''(f) \geq 0$, $g(f)$ attains its maximum at some $f' \in [0.5, 1)$ satisfying $g'(f') = 0$, i.e.,

$$\begin{aligned} 0 &= \frac{2st}{\phi} + \frac{2k(1-2f')}{s} \\ f' &= \frac{1}{2} + \frac{s^2 t}{2k\phi}. \end{aligned} \tag{B.6.5}$$

Then

$$\begin{aligned}\max_{f\in[0.5,1)} g(f) = g(f') &= -\frac{1}{2} + \frac{st(2f'-1)}{\phi} + \frac{2kf'(1-f')}{s} \\ &= -\frac{1}{2} + \frac{s^3t^2}{k\phi^2} + \frac{2k}{s}\left(\frac{1}{4} - \frac{s^4t^2}{4k^2\phi^2}\right) \\ &= -\frac{1}{2} + \frac{s^3t^2}{2k\phi^2} + \frac{k}{2s} \\ &< -\frac{1}{2} + \frac{s}{2k}\left(\frac{k}{s}\right)^2 + \frac{k}{2s} \\ &= -\frac{1}{2} + \frac{k}{s} < 0 \text{ if } s > 2k.\end{aligned} \tag{B.6.6}$$

Across both scenarios, $\frac{\partial CS^*}{\partial k} \leq g(f) < 0$ if $s > 2k$ and $s > \frac{c^2+d^2+4cd+2m(c+d)+m^2}{4t}$. Accordingly, when k increases, the equilibrium consumer surplus decreases whenever the degree of consumers' horizontal differentiation is sufficiently large relative to consumers' switching costs and the magnitudes of users' network effects. □

B.7 Proof of Proposition 6

B.7.1 Part 1

From (4.2.3) and (B.1.5),

$$\begin{aligned}\frac{\partial DS^*}{\partial k} &= \frac{t}{2}\frac{(c+d+m)(2f-1)\times 2[(n_0^{D*}+n_2^{D*})-(n_1^{D*}+n_2^{D*})]}{2\phi} \\ &= \frac{t(c+d+m)(2f-1)(n_0^{D*}-n_1^{D*})}{2\phi}.\end{aligned} \tag{B.7.1}$$

With $f \in [0.5, 1)$, which indicates that platform 0 is (weakly) strong on the consumer side, $\frac{\partial DS^*}{\partial k} > 0$ if and only if $f > 0.5$ and $n_0^{D*} - n_1^{D*} > 0$. Accordingly, the equilibrium developer surplus increases in consumers' switching costs if and only if the strong platform serves more developers in the equilibrium than the weak platform.

Similarly, and in addition, $\frac{\partial DS^*}{\partial k} = 0$ if and only if $f = 0.5$ or $n_0^{D*} = n_1^{D*}$. In other words, the equilibrium developer surplus is invariant to any change in consumers' switching costs when neither platform is strong or the platforms serve the same number of developers in the equilibrium. □

B.7.2 Part 2

Because (A.2.4) ensures $n_0^{D*}, n_1^{D*} \in (0,1)$, and since $f \in (0,1)$, from (B.7.1)

$$\begin{aligned}\left|\frac{\partial DS^*}{\partial k}\right| &= \frac{t(c+d+m)|2f-1||n_0^{D*}-n_1^{D*}|}{2\phi} \\ &< \frac{t(c+d+m)}{2\phi}.\end{aligned} \tag{B.7.2}$$

Accordingly, for any $\epsilon > 0$, $\left|\frac{\partial DS^*}{\partial k}\right| < \epsilon$ whenever $\frac{t(c+d+m)}{2\phi} < \epsilon$, i.e.,

$$\frac{1}{6t}\Big[\frac{t(c+d+m)}{2\epsilon} + c^2 + d^2 + 4cd + 2m(c+d) + m^2\Big] < s. \tag{B.7.3}$$

Define $s_\epsilon := \frac{1}{6t}\Big[\frac{t(c+d+m)}{2\epsilon} + c^2 + d^2 + 4cd + 2m(c+d) + m^2\Big]$. Then, from (B.7.3), for any small $\epsilon > 0$, $\left|\frac{\partial DS^*}{\partial k}\right| < \epsilon$ whenever $s > s_\epsilon$. In other words, as consumers become sufficiently horizontally differentiated, the change in the equilibrium developer surplus in response to any change in k diminishes in magnitude and ultimately vanishes. □

B.8 Proof of Proposition 7

B.8.1 Part 1

From (4.2.2), (B.2.1), (B.2.3), and (B.2.4),

$$\frac{\partial CS^*}{\partial r} = c(-\frac{1}{2t} + \frac{1}{t}) + \frac{d+m}{2t} = \frac{c+d+m}{2t} > 0. \tag{B.8.1}$$

Accordingly, the equilibrium consumer surplus increases in r. □

B.8.2 Part 2

From (4.2.3) and (B.2.4),

$$\begin{aligned}\frac{\partial DS^*}{\partial r} &= \frac{t}{2} \times 2[(n_0^{D*} + n_2^{D*})(-\frac{1}{2t} + \frac{1}{t}) + (n_1^{D*} + n_2^{D*})(-\frac{1}{2t} + \frac{1}{t})] - 1 \\ &= -\frac{1-n_2^{D*}}{2} < 0 \text{ by (A.2.4)}.\end{aligned} \tag{B.8.2}$$

Accordingly, the equilibrium developer surplus decreases in r. □

C Mathematical Content Omitted in Section 5

C.1 Steps for Solving the Extension with Platforms' Internal Diversity

Section 5 first extends the baseline model by allowing the platforms' locations to be l_0 and l_1 with $0 \le l_0 < l_1 \le 1$.[48] In this extension, the equilibrium prices are not necessarily directly obtainable from the first-order conditions of the platforms' profit maximization problems. The reason is that, in some cases, following the argument of d'Aspremont et al. (1979), platforms' profit functions may be discontinuous at certain price levels where an entire group of users is indifferent between the two platforms. As a result, an equilibrium does not always exist; it exists when all indifferent users lie between the two platforms on the Hotelling line.

The players' payoffs in this extension are as follows.

- For a consumer with $(x, b) \in [0,1]\times\{0,1\}$ using platform $i \in \{0,1\}$, the payoff is $\tilde{U}_i(x,b)$, where

$$\begin{aligned} \tilde{U}_0(x,b) &= u_0 - s|x - l_0| + c(n_0^D + n_2^D) - p_0 - k\mathbb{1}[b=1], \\ \tilde{U}_1(x,b) &= u_1 - s|x - l_1| + c(n_1^D + n_2^D) - p_1 - k\mathbb{1}[b=0]. \end{aligned} \tag{C.1.1}$$

- For a single-homing developer with $y \in [0,1]$ operating on platform $i \in \{0,1\}$, the payoff is $\tilde{V}_i(y)$, where

$$\begin{aligned} \tilde{V}_0(y) &= v_0 - t|y - l_0| + dn_0^C - q_0, \\ \tilde{V}_1(y) &= v_1 - t|y - l_1| + dn_1^C - q_1. \end{aligned} \tag{C.1.2}$$

- For a multi-homer with $y \in [0,1]$, the payoff is

$$\tilde{V}_2(y) = \tilde{V}_0(y) + \tilde{V}_1(y) + r. \tag{C.1.3}$$

- For each platform $i \in \{0,1\}$, the payoff is its profit, the expression for which is identical to that in (2.4):

$$\tilde{\pi}_i = p_i n_i^C + (q_i + m n_i^C)(n_i^D + n_2^D). \tag{C.1.4}$$

[48] This range of parameters includes the baseline setting of Hotelling (1929) and excludes the case with $l_0 = l_1$ in which the game degenerates into Bertrand competition.

Accordingly, I find an equilibrium $[(\tilde{p}_0^*, \tilde{q}_0^*), (\tilde{p}_1^*, \tilde{q}_1^*), \tilde{L}_C^*, \tilde{L}_D^*]$ with market shares $(\tilde{n}_i^{C*})_{i=0}^1$ and $(\tilde{n}_i^{D*})_{i=0}^2$ by first solving for $(\tilde{n}_i^{C*})_{i=0}^1$ and $(\tilde{n}_i^{D*})_{i=0}^2$ from the second stage of the game.[49]

At the second stage, because platforms' prices $(p_i, q_i)_{i=0}^1$ are given, denote consumers' cutoff values by $\tilde{x}_0^*$ and $\tilde{x}_1^*$ so that the equilibrium strategy of a consumer with $(x, b) \in [0,1] \times \{0,1\}$ is

$$\tilde{L}_C^*(x, b, p_0, q_0, p_1, q_1) = \begin{cases} 0 & \text{if } x \le \tilde{x}_b^* \\ 1 & \text{elsewhere} \end{cases}. \tag{C.1.5}$$

Accordingly, for each $b \in \{0,1\}$, $\tilde{U}_0(\tilde{x}_b^*, b) = \tilde{U}_1(\tilde{x}_b^*, b)$. With $\tilde{x}_0^*, \tilde{x}_1^* \in [l_0, l_1]$, which is needed for equilibrium existence, (C.1.1) implies that $\tilde{U}_0(\tilde{x}_0^*, 0) = \tilde{U}_1(\tilde{x}_0^*, 0)$ is equivalent to

$$\begin{aligned} u_0 - s(x - l_0) + c(\tilde{n}_0^{D*} + \tilde{n}_2^{D*}) - p_0 = u_1 - s(l_1 - x) + c(\tilde{n}_1^{D*} + \tilde{n}_2^{D*}) - p_1 - k, \text{ i.e.,} \\ \tilde{x}_0^* = \frac{u_0 + sl_0 - u_1 + sl_1 - p_0 + p_1 + c(\tilde{n}_0^{D*} - \tilde{n}_1^{D*}) + k}{2s}. \end{aligned} \tag{C.1.6}$$

Similarly, $\tilde{U}_0(\tilde{x}_1^*, 1) = \tilde{U}_1(\tilde{x}_1^*, 1)$ is equivalent to

$$\begin{aligned} u_0 - s(x - l_0) + c(\tilde{n}_0^{D*} + \tilde{n}_2^{D*}) - p_0 - k = u_1 - s(l_1 - x) + c(\tilde{n}_1^{D*} + \tilde{n}_2^{D*}) - p_1, \text{ i.e.,} \\ \tilde{x}_1^* = \frac{u_0 + sl_0 - u_1 + sl_1 - p_0 + p_1 + c(\tilde{n}_0^{D*} - \tilde{n}_1^{D*}) - k}{2s}. \end{aligned} \tag{C.1.7}$$

Because consumers' horizontal preferences are independent of the platforms they initially patronize, the resulting consumer-side market shares are

$$\begin{aligned} \tilde{n}_0^{C*} &= f\tilde{x}_0^* + (1-f)\tilde{x}_1^* \\ &= \frac{u_0 + sl_0 - u_1 + sl_1 - p_0 + p_1 + c(\tilde{n}_0^{D*} - \tilde{n}_1^{D*}) + k(2f-1)}{2s} \\ &= \frac{1}{2} + \frac{\tilde{u}_- - p_0 + p_1 + c(\tilde{n}_0^{D*} - \tilde{n}_1^{D*})}{2s}, \\ \tilde{n}_1^{C*} &= 1 - \tilde{n}_0^{C*} = \frac{1}{2} - \frac{\tilde{u}_- - p_0 + p_1 + c(\tilde{n}_0^{D*} - \tilde{n}_1^{D*})}{2s}, \end{aligned} \tag{C.1.8}$$

where $\tilde{u}_- := u_0 + sl_0 - u_1 - s(1 - l_1) + k(2f - 1)$.

Similarly, denote developers' cutoff values by $\tilde{n}_0^{D*}$ and $1 - \tilde{n}_1^{D*}$. The equilibrium strategy

[49] In such an equilibrium, the indifferent consumers are located at $\tilde{x}_0^*$ and $\tilde{x}_1^*$ (see (C.1.6) and (C.1.7)), and the indifferent developers are located at $\tilde{n}_0^{D*}$ and $1 - \tilde{n}_1^{D*}$ (see (C.1.10) and (C.1.11)).

of a developer with $y \in [0,1]$ is

$$\tilde{L}_D^*(y, p_0, q_0, p_1, q_1) = \begin{cases} 0 & \text{if } y < \tilde{n}_0^{D*} \\ 1 & \text{if } y > 1 - \tilde{n}_1^{D*} \\ 2 & \text{elsewhere} \end{cases}. \tag{C.1.9}$$

Accordingly, $\tilde{V}_0(\tilde{n}_0^{D*}) = \tilde{V}_2(\tilde{n}_0^{D*})$. and $\tilde{V}_1(1 - \tilde{n}_1^{D*}) = \tilde{V}_2(1 - \tilde{n}_1^{D*})$. Combining (C.1.2) and (C.1.3) with $\tilde{n}_0^{D*}, 1 - \tilde{n}_1^{D*} \in [l_0, l_1]$, which is needed for equilibrium existence, $\tilde{V}_0(\tilde{n}_0^{D*}) = \tilde{V}_2(\tilde{n}_0^{D*})$ is equivalent to

$$\begin{aligned} 0 &= \tilde{V}_1(\tilde{n}_0^{D*}) + r \\ &= v_1 - t(l_1 - \tilde{n}_0^{D*}) + d\tilde{n}_1^{C*} - q_1 + r, \text{ i.e.,} \\ \tilde{n}_0^{D*} &= l_1 - \frac{v_1 - q_1 + d\tilde{n}_1^{C*} + r}{t} \\ &= 1 - \frac{v_1 + t(1 - l_1) - q_1 + d\tilde{n}_1^{C*} + r}{t}. \end{aligned} \tag{C.1.10}$$

Similarly, $\tilde{V}_1(1 - \tilde{n}_1^{D*}) = \tilde{V}_2(1 - \tilde{n}_1^{D*})$ is equivalent to

$$\begin{aligned} 0 &= \tilde{V}_0(1 - \tilde{n}_1^{D*}) + r \\ &= v_0 - t(1 - \tilde{n}_1^{D*} - l_0) + d\tilde{n}_0^{C*} - q_0 + r, \text{ i.e.,} \\ \tilde{n}_1^{D*} &= 1 - l_0 - \frac{v_0 - q_0 + d\tilde{n}_0^{C*} + r}{t} \\ &= 1 - \frac{v_0 + tl_0 - q_0 + d\tilde{n}_0^{C*} + r}{t}, \text{ and, by definition,} \\ \tilde{n}_2^{D*} &= 1 - \tilde{n}_0^{D*} - \tilde{n}_1^{D*}. \end{aligned} \tag{C.1.11}$$

Finally, combining (C.1.8), (C.1.10), and (C.1.11) yields a system of five equations for

five unknowns. The solution to the equation system is

$$\begin{aligned}
\tilde{n}_0^{C*} &= \frac{1}{2} + \frac{c(\tilde{v}_- - q_0 + q_1) + t(\tilde{u}_- - p_0 + p_1)}{2(st - cd)}, \\
\tilde{n}_1^{C*} &= \frac{1}{2} - \frac{c(\tilde{v}_- - q_0 + q_1) + t(\tilde{u}_- - p_0 + p_1)}{2(st - cd)}, \\
\tilde{n}_0^{D*} &= 1 - \frac{\tilde{v}_+ - q_0 - q_1}{2t} + \frac{s(\tilde{v}_- - q_0 + q_1) + d(\tilde{u}_- - p_0 + p_1)}{2(st - cd)}, \\
\tilde{n}_1^{D*} &= 1 - \frac{\tilde{v}_+ - q_0 - q_1}{2t} - \frac{s(\tilde{v}_- - q_0 + q_1) + d(\tilde{u}_- - p_0 + p_1)}{2(st - cd)}, \\
\tilde{n}_2^{D*} &= \frac{\tilde{v}_+ - q_0 - q_1}{t} - 1,
\end{aligned} \tag{C.1.12}$$

where $\tilde{u}_- := u_0 + sl_0 - u_1 - s(1 - l_1) + k(2f - 1)$, $\tilde{v}_- := v_0 + tl_0 - v_1 - t(1 - l_1)$, and $\tilde{v}_+ := v_0 + tl_0 + v_1 + t(1 - l_1) + d + 2r$.

Accordingly, the only difference between (C.1.12) and (A.1.4) is the replacement of u_-, v_-, v_+ by $\tilde{u}_-, \tilde{v}_-, \tilde{v}_+$. Thus, in this extension, following the same procedure as in Appendix A.1, the equilibrium prices, market shares, and strategies of users retain the same form as in (3.1)–(3.4). The only changes arise from substituting $\tilde{u}_-, \tilde{v}_-, \tilde{v}_+$ for u_-, v_-, v_+. Therefore, the results in Section 4 remain valid under this extension, and the corresponding proofs follow directly from the arguments in Appendix B.

C.2 Steps for Solving the Extension with Platforms' Price Rigidity on the Developer Side

In this second extension, I find an equilibrium $(\hat{p}_0^*, \hat{p}_1^*, \hat{L}_C^*, \hat{L}_D^*)$ with market shares $(\hat{n}_i^{C*})_{i=0}^1$ and $(\hat{n}_i^{D*})_{i=0}^2$ by solving the model from the second stage of the game.

First, given users' optimal platform choices, $(\hat{n}_i^{C*})_{i=0}^1 = (n_i^{C*})_{i=0}^1$ and $(\hat{n}_i^{D*})_{i=0}^2 = (n_i^{D*})_{i=0}^2$, where $(n_i^{C*})_{i=0}^1$ and $(n_i^{D*})_{i=0}^2$ are defined in (A.1.4). The relevant derivation is identical to the corresponding part of Appendix A.1.

Next, I solve for $\hat{p}_0^*$ and $\hat{p}_1^*$ from the platforms' profit maximization problems. In this extension, for each platform $i \in \{0, 1\}$, the problem is

$$\max_{p_i} \hat{\pi}_i = p_i \hat{n}_i^{C*} + (q_i + m\hat{n}_i^{C*})(\hat{n}_i^{D*} + \hat{n}_2^{D*}), \tag{C.2.1}$$

where q_i is exogenous.

The first-order condition is

$$\frac{\partial \hat{\pi}_i}{\partial p_i} = \hat{n}_i^{C*} - \frac{t[p_i + m(\hat{n}_i^{D*} + \hat{n}_2^{D*})]}{2(st - cd)} - \frac{d(q_i + m\hat{n}_i^{C*})}{2(st - cd)} = 0, \tag{C.2.2}$$

and the second-order condition is

$$\frac{\partial^2 \hat{\pi}_i}{\partial p_i^2} = -\frac{t[2(st - cd) - dm]}{2(st - cd)^2} < 0, \tag{C.2.3}$$

for which (A.1.9) is a sufficient condition.

Under (A.1.9), the first-order condition in (C.2.2) yields a unique profit-maximizing price for each platform i. Combining the two first-order conditions for the platforms yields the equilibrium prices shown in (5.1), which constitute the unique equilibrium in this extension whenever the interior solution in (5.1) is valid. Similar to Appendix A.2, a set of sufficient conditions ensuring this validity includes (A.1.9) and the following four conditions:

1. Consumers' participation constraint.

   In equilibrium, every consumer prefers using a platform to not using any platform. Equivalently, $U_0(\hat{x}_b^*, b) = U_1(\hat{x}_b^*, b) \geq 0$ for each $b \in \{0, 1\}$,[50] which holds under

   $$u_0 + u_1 - k - s + \frac{(c + m)v_+ - (c - d + m)(q_0 + q_1) - 2(st - cd) + dm}{t} \geq 0. \tag{C.2.4}$$

2. Developers' participation constraint.

   In equilibrium, every developer prefers operating on at least one platform to operating on neither.[51] Equivalently, $V_0(\hat{n}_0^{D*}) = V_2(\hat{n}_0^{D*}) \geq 0$ and $V_1(1 - \hat{n}_1^{D*}) = V_2(1 - \hat{n}_1^{D*}) \geq 0$, which hold under

   $$v_+ - r - t - q_0 - q_1 \geq 0. \tag{C.2.5}$$

3. Interior consumer-side market shares.

   The equilibrium satisfies $\hat{x}_0^*, \hat{x}_1^* \in (0, 1)$, where $\hat{x}_0^* = \hat{n}_0^{C*} + \frac{k(1-f)}{s}$ and $\hat{x}_1^* = \hat{n}_0^{C*} - \frac{kf}{s}$ are counterparts of x_0^* and x_1^* defined in Section 3 for (3.3). This means that each platform serves some consumers that initially patronize itself and serves some consumers that

[50] Analogous to Section 3, $\hat{x}_0^* = \hat{n}_0^{C*} + \frac{k(1-f)}{s}$ and $\hat{x}_1^* = \hat{n}_0^{C*} - \frac{kf}{s}$.
[51] Because $r \geq 0$, this condition also guarantees $\hat{n}_2^{D*} = \frac{v_+ - q_0 - q_1 - t}{t} \geq 0$.

initially patronize its rival. This implies $\hat{n}_0^{C*}, \hat{n}_1^{C*} \in (0,1)$ and holds under

$$-1 + \frac{2kf}{s} < \left| \frac{(c+m)v_- + tu_- - (c-d+m)(q_0-q_1)}{\hat{\phi}} \right| < 1 - \frac{2k(1-f)}{s}. \tag{C.2.6}$$

4. Interior developer-side market shares.

   The equilibrium satisfies $\hat{n}_0^{D*}, \hat{n}_1^{D*} \in (0,1)$, meaning that each platform serves some single-homing developers.[52] This holds under

$$\left| \frac{(3st - 2cd - dm)v_- + tdu_- - (3st - 2cd - d^2 - dm)(q_0 - q_1)}{\hat{\phi}} \right| < 2t - v_+ + q_0 + q_1. \tag{C.2.7}$$

C.3 Omitted Proofs

C.3.1 Proof of Proposition 1

Part 1.

From (5.1), with $u_- = u_0 - u_1 + k(2f-1)$,

$$\frac{\partial \hat{p}_0^*}{\partial k} = -\frac{\partial \hat{p}_1^*}{\partial k} = \frac{(st - cd - dm)(2f-1)}{\hat{\phi}}. \tag{C.3.1.1}$$

Under (A.1.9), **!!!**

$$\begin{aligned} st - cd - dm &> st - (s - \frac{m}{2})(t - \frac{m}{2}) - dm \\ &= \frac{m}{2}(s + t - 2d) \geq 0, \\ \hat{\phi} &= 3(st - cd) - 2dm > st - cd > 0. \end{aligned} \tag{C.3.1.2}$$

Therefore, combining (C.3.1.1) and (C.3.1.2) yields $\frac{\partial \hat{p}_0^*}{\partial k} = -\frac{\partial \hat{p}_1^*}{\partial k} \geq 0$, where the weak inequality binds if and only if $f = 0.5$. Accordingly, when k increases, a platform that is strong (weak) on the consumer side charges a higher (lower) price to consumers. □

Part 2.

Because q_0 and q_1 are assumed to be fixed, this part is not applicable in this extension.

Part 3.

[52] This condition implicitly requires $\hat{n}_2^{D*} = \frac{v_+ - q_0 - q_1 - t}{t} < 1$.

From (5.2), with $u_- = u_0 - u_1 + k(2f-1)$,

$$\frac{\partial \hat{n}_0^{C*}}{\partial k} = -\frac{\partial \hat{n}_1^{C*}}{\partial k} = \frac{t(2f-1)}{2\hat{\phi}} \geq 0 \text{ by (C.3.1.2)}, \tag{C.3.1.3}$$

and the weak inequality binds if and only if $f = 0.5$. Accordingly, when k increases, a platform that is strong (weak) on the consumer side serves more (fewer) consumers. □

Part 4.

From (5.2), $\frac{\partial \hat{n}_2^{D*}}{\partial k} = 0$. Also, combining (5.2) with $u_- = u_0 - u_1 + k(2f-1)$ yields

$$\frac{\partial \hat{n}_0^{D*}}{\partial k} = -\frac{\partial \hat{n}_1^{D*}}{\partial k} = \frac{d(2f-1)}{2\hat{\phi}} \geq 0 \text{ by (C.3.1.2)}, \tag{C.3.1.4}$$

where the weak inequality binds if and only if $f = 0.5$. Accordingly, when k increases, a platform that is strong (weak) on the consumer side serves more (fewer) developers, and the number of multi-homers remains unchanged. □

C.3.2 Proof of Proposition 2

Part 1.

From (5.1), with $v_+ = v_0 + v_1 + d + 2r$,

$$\frac{\partial \hat{p}_0^*}{\partial r} = \frac{\partial \hat{p}_1^*}{\partial r} = -\frac{2m}{2t} = -\frac{m}{t} \leq 0. \tag{C.3.2.1}$$

Accordingly, because $m \geq 0$, when r increases, each platform charges a lower price to developers unless the commission m is 0. Under this extension, this part is quantitatively different from the corresponding part in the baseline model, where each platform always charges a lower price to consumers when r increases even if $m = 0$. □

Part 2.

Because q_0 and q_1 are assumed to be fixed, this part is not applicable in this extension.

Part 3.

From (5.2),

$$\frac{\partial \hat{n}_0^{C*}}{\partial r} = \frac{\partial \hat{n}_1^{C*}}{\partial r} = 0. \tag{C.3.2.2}$$

Accordingly, when r increases, each platform still serves the same number of consumers. □

Part 4.

From (5.2), with $v_+ = v_0 + v_1 + d + 2r$,

$$\begin{aligned} \frac{\partial \hat{n}_0^{D*}}{\partial r} &= \frac{\partial \hat{n}_1^{D*}}{\partial r} = -\frac{2}{2t} = -\frac{1}{t} < 0, \\ \frac{\partial \hat{n}_2^{D*}}{\partial r} &= \frac{2}{t} = \left|\frac{\partial \hat{n}_0^{D*}}{\partial r}\right| + \left|\frac{\partial \hat{n}_1^{D*}}{\partial r}\right| > 0. \end{aligned} \tag{C.3.2.3}$$

Accordingly, when r increases, each platform serves more developers, and on each platform, some developers switch from single-homing to multi-homing. □

C.3.3 Proof of Proposition 3

First, similar to (4.2.1), in this extension, the equilibrium profit of platform $i \in \{0, 1\}$ is

$$\hat{\pi}_i^* = \hat{p}_i^* \hat{n}_i^{C*} + (q_i + m\hat{n}_i^{C*})(\hat{n}_i^{D*} + \hat{n}_2^{D*}). \tag{C.3.3.1}$$

From (C.3.1.1), (C.3.1.3), (C.3.1.4), and (C.3.3.1),

$$\begin{aligned} \frac{\partial \hat{\pi}_0^*}{\partial k} &= \frac{(2f-1)\{(st-cd-dm)\hat{n}_0^{C*} + t\hat{p}_0^* + d(q_0 + m\hat{n}_0^{C*}) + tm(\hat{n}_0^{D*} + \hat{n}_2^{D*})\}}{2\hat{\phi}} \\ &= \frac{(2f-1)[2(st-cd)-dm]\hat{n}_0^{C*}}{\hat{\phi}}. \end{aligned} \tag{C.3.3.2}$$

Because (1) (C.2.6) implies that $\hat{n}_0^{C*} > 0$ and (2) combining of (C.3.1.2) and (A.1.9) implies that $2(st-cd) - dm > st - cd > 0$, it follows that $\frac{\partial \hat{\pi}_0^*}{\partial k} \geq 0$, where the weak inequality binds if and only if $f = 0.5$. Symmetrically,

$$\begin{aligned} \frac{\partial \hat{\pi}_1^*}{\partial k} &= \frac{-(2f-1)\{(st-cd-dm)\hat{n}_1^{C*} + t\hat{p}_1^* + d(q_1 + m\hat{n}_1^{C*}) + tm(\hat{n}_1^{D*} + \hat{n}_2^{D*})\}}{2\hat{\phi}} \\ &= \frac{-(2f-1)[2(st-cd)-dm]\hat{n}_1^{C*}}{\hat{\phi}} \leq 0, \end{aligned} \tag{C.3.3.3}$$

and the weak inequality binds if and only if $f = 0.5$.

Accordingly, when k increases, the equilibrium profit of a platform increases (decreases) if and only if the platform is strong (weak) on the consumer side. □

C.3.4 Proof of Proposition 8

In the second extension introduced in Section 5, Proposition 8 serves as the counterpart of Proposition 4 in the baseline model.

From (C.3.2.1)–(C.3.2.3) and (C.3.3.1),

$$\begin{aligned}\frac{\partial \hat{\pi}_0^*}{\partial r} &= -\frac{m}{t}\hat{n}_0^{C*} + (q_0 + m\hat{n}_0^{C*})(-\frac{1}{t} + \frac{2}{t}) = \frac{q_0}{t},\\ \frac{\partial \hat{\pi}_1^*}{\partial r} &= -\frac{m}{t}\hat{n}_1^{C*} + (q_1 + m\hat{n}_1^{C*})(-\frac{1}{t} + \frac{2}{t}) = \frac{q_1}{t}.\end{aligned} \tag{C.3.4.1}$$

Accordingly, when r increases, the equilibrium profit of a platform increases (decreases) if and only if it charges a positive (negative) price to developers. □

C.3.5 Proof of Part 1 of Proposition 5

First of all, similar to (4.2.2), the equilibrium consumer surplus in this extension is

$$\begin{aligned}\hat{CS}^* &= f\Big[\int_0^{\hat{x}_0^*} \hat{U}_0^*(x,0)dx + \int_{\hat{x}_0^*}^1 \hat{U}_1^*(x,0)dx\Big] + (1-f)\Big[\int_0^{\hat{x}_1^*} \hat{U}_0^*(x,1)dx + \int_{\hat{x}_1^*}^1 \hat{U}_1^*(x,1)dx\Big]\\ &= u_1 + c(\hat{n}_1^{D*} + \hat{n}_2^{D*}) - \hat{p}_1^* + s\hat{n}_0^{C*2} - fk + \frac{k^2 f(1-f)}{s} - \frac{s}{2},\end{aligned} \tag{C.3.5.1}$$

where $\hat{U}_i^*(x,b)$ is calculated in accordance with (2.1) and denotes the equilibrium payoff of a consumer with $x \in [0,1]$, initially patronizing platform $b \in \{0,1\}$, and using platform $i \in \{0,1\}$ in this extension.

From (C.3.1.1), (C.3.1.3), (C.3.1.4), and (C.3.5.1), and because (C.2.6) gives $\hat{n}_0^{C*} < 1$, for any $f \in [0.5, 1)$,

$$\begin{aligned}\frac{\partial \hat{CS}^*}{\partial k} &= \frac{[-cd + 2(st - cd - dm)](2f-1)}{2\hat{\phi}} + \frac{2st\hat{n}_0^{C*}(2f-1)}{2\hat{\phi}} - f + \frac{2kf(1-f)}{s}\\ &\le \frac{(2st - 3cd - 2dm)(2f-1)}{2\hat{\phi}} + \frac{st(2f-1)}{\hat{\phi}} - f + \frac{2kf(1-f)}{s}\\ &= -\frac{1}{2} + \frac{st(2f-1)}{2\hat{\phi}} + \frac{2kf(1-f)}{s}.\end{aligned} \tag{C.3.5.2}$$

Define $\hat{g}(f) := -\frac{1}{2} + \frac{st(2f-1)}{2\hat{\phi}} + \frac{2kf(1-f)}{s}$ for all $f \in [0.5, 1)$, then

$$\begin{aligned}\hat{g}'(f) &= \frac{st}{\hat{\phi}} + \frac{2k(1-2f)}{s},\\ \hat{g}''(f) &= -\frac{4k}{s} \le 0.\end{aligned} \tag{C.3.5.3}$$

From (C.3.1.2), $\hat{g}'(0.5) = \frac{st}{\hat{\phi}} > 0$. Thus, to determine the maximum value of $\hat{g}(f)$ on

$[0.5, 1)$, I consider the following two scenarios.

Scenario 1: $\hat{g}'(1) \geq 0$

Since $\hat{g}''(f) = -\frac{4k}{s} \leq 0$, $\hat{g}'(f) \geq 0$ for all $f \in [0.5, 1)$. Therefore,

$$\max_{f \in [0.5,1)} \hat{g}(f) = \hat{g}(1) = -\frac{1}{2} + \frac{st}{2\hat{\phi}} < 0 \text{ if and only if } st < \hat{\phi}, \tag{C.3.5.4}$$

i.e., $s > \frac{\frac{3}{2}cd+dm}{t}$.

Scenario 2: $\hat{g}'(1) < 0$

In this scenario,

$$0 > \hat{g}'(1) = \frac{2st}{\hat{\phi}} - \frac{2k}{s}, \tag{C.3.5.5}$$

i.e., $\frac{k}{s} > \frac{st}{\hat{\phi}}$.

Since $\hat{g}'(0.5) > 0$ and $\hat{g}''(f) \geq 0$, $\hat{g}(f)$ attains its maximum at some $\hat{f}' \in [0.5, 1)$ satisfying $\hat{g}'(\hat{f}') = 0$, i.e.,

$$\begin{aligned} 0 &= \frac{2st}{\hat{\phi}} + \frac{2k(1-2\hat{f}')}{s} \\ \hat{f}' &= \frac{1}{2} + \frac{s^2 t}{2k\hat{\phi}}. \end{aligned} \tag{C.3.5.6}$$

Then

$$\begin{aligned} \max_{f \in [0.5,1)} \hat{g}(f) = \hat{g}(\hat{f}') &= -\frac{1}{2} + \frac{st(2\hat{f}'-1)}{\hat{\phi}} + \frac{2k\hat{f}'(1-\hat{f}')}{s} \\ &= -\frac{1}{2} + \frac{s^3 t^2}{k\hat{\phi}^2} + \frac{2k}{s}\Big(\frac{1}{4} - \frac{s^4 t^2}{4k^2\hat{\phi}^2}\Big) \\ &= -\frac{1}{2} + \frac{s^3 t^2}{2k\hat{\phi}^2} + \frac{k}{2s} \\ &< -\frac{1}{2} + \frac{s}{2k}\Big(\frac{k}{s}\Big)^2 + \frac{k}{2s} \\ &= -\frac{1}{2} + \frac{k}{s} < 0 \text{ if } s > 2k. \end{aligned} \tag{C.3.5.7}$$

Across both scenarios, $\frac{\partial \hat{CS}^*}{\partial k} \leq \hat{g}(f) < 0$ if $s > 2k$ and $s > \frac{\frac{3}{2}cd+dm}{t}$. Accordingly, when k increases, the equilibrium consumer surplus decreases whenever the degree of consumers' horizontal differentiation is sufficiently large relative to consumers' switching costs and the

magnitudes of users' network effects. Under this extension, this part is quantitatively different from the corresponding part in the baseline model, where the degree of consumers' horizontal differentiation is sufficiently large if $s > 2k$ and $s > \frac{c^2+d^2+4cd+2m(c+d)+m^2}{4t}$, rather than $s > 2k$ and $s > \frac{\frac{3}{2}cd+dm}{t}$. □

C.3.6 Proof of Proposition 6

Part 1.

First, similar to (4.2.3), the equilibrium developer surplus in this extension is

$$\begin{aligned} \hat{DS}^* &= \int_0^{\hat{n}_0^{D*}} \hat{V}_0^*(y)dy + \int_{\hat{n}_0^{D*}}^{\hat{n}_0^{D*}+\hat{n}_2^{D*}} \hat{V}_2^*(y)dy + \int_{\hat{n}_0^{D*}+\hat{n}_2^{D*}}^{1} \hat{V}_1^*(y)dy \\ &= \frac{t}{2}[(\hat{n}_0^{D*}+\hat{n}_2^{D*})^2 + (\hat{n}_1^{D*}+\hat{n}_2^{D*})^2] - r, \end{aligned} \tag{C.3.6.1}$$

where $\hat{V}_i^*(y)$ is calculated in accordance with (2.2) and (2.3), and it denotes the equilibrium payoff of a developer with $y \in [0,1]$, operating on platform i (when $i \in \{0,1\}$) or multi-homing (when $i = 2$) in this extension.

From (C.3.1.4) and (C.3.6.1),

$$\begin{aligned} \frac{\partial \hat{DS}^*}{\partial k} &= \frac{t}{2}\frac{d(2f-1)\times 2[(\hat{n}_0^{D*}+\hat{n}_2^{D*}) - (\hat{n}_1^{D*}+\hat{n}_2^{D*})]}{2\hat{\phi}} \\ &= \frac{td(2f-1)(\hat{n}_0^{D*}-\hat{n}_1^{D*})}{2\hat{\phi}}. \end{aligned} \tag{C.3.6.2}$$

With $f \in [0.5, 1)$, which indicates that platform 0 is (weakly) strong on the consumer side, $\frac{\partial \hat{DS}^*}{\partial k} > 0$ if and only if $f > 0.5$ and $\hat{n}_0^{D*} - \hat{n}_1^{D*} > 0$. Accordingly, the equilibrium developer surplus increases in consumers' switching costs if and only if the strong platform serves more developers in the equilibrium than the weak platform.

Similarly, and in addition, $\frac{\partial \hat{DS}^*}{\partial k} = 0$ if and only if $f = 0.5$ or $\hat{n}_0^{D*} = \hat{n}_1^{D*}$. In other words, the equilibrium developer surplus is invariant to any change in consumers' switching costs when neither platform is strong or the platforms serve the same number of developers in the equilibrium. □

Part 2.

Because (C.2.7) ensures $\hat{n}_0^{D*}, \hat{n}_1^{D*} \in (0,1)$, and since $f \in (0,1)$,

$$\left|\frac{\partial \hat{DS}^*}{\partial k}\right| = \frac{td|2f-1||\hat{n}_0^{D*}-\hat{n}_1^{D*}|}{2\hat{\phi}} < \frac{td}{2\hat{\phi}}. \tag{C.3.6.3}$$

Accordingly, for any $\epsilon > 0$, $\left|\frac{\partial \hat{DS}^*}{\partial k}\right| < \epsilon$ whenever $\frac{td}{2\hat{\phi}} < \epsilon$, i.e.,

$$\frac{1}{3t}\Big(\frac{td}{2\epsilon} + 3cd + 2dm\Big) < s. \tag{C.3.6.4}$$

Define $\hat{s}_\epsilon := \frac{1}{3t}\Big(\frac{td}{2\epsilon} + 3cd + 2dm\Big)$. Then, from (C.3.6.4), for any small $\epsilon > 0$, $\left|\frac{\partial \hat{DS}^*}{\partial k}\right| < \epsilon$ whenever $s > \hat{s}_\epsilon$. In other words, as consumers become sufficiently horizontally differentiated, the change in the equilibrium developer surplus in response to any change in k diminishes in magnitude and ultimately vanishes. □

C.3.7 Proofs of Propositions 7 and 9

Proof of Part 1 of Proposition 7.

From (C.3.2.1)–(C.3.2.3) and (C.3.5.1),

$$\frac{\partial \hat{CS}^*}{\partial r} = c\big(-\frac{1}{t} + \frac{2}{t}\big) + \frac{m}{t} = \frac{c+m}{t} > 0. \tag{C.3.7.1}$$

Accordingly, the equilibrium consumer surplus increases in r. □

Proof of Proposition 9.

In the second extension introduced in Section 5, Proposition 9 serves as the counterpart of part 2 of Proposition 7 in the baseline model.

From (C.3.2.3) and (C.3.6.1),

$$\frac{\partial \hat{DS}^*}{\partial r} = \frac{t}{2} \times 2(\hat{n}_0^{D*} + \hat{n}_2^{D*} + \hat{n}_1^{D*} + \hat{n}_2^{D*}) \times \frac{1}{t} - 1 = \hat{n}_2^{D*} > 0 \text{ by (C.2.6)}. \tag{C.3.7.2}$$

Accordingly, the equilibrium developer surplus increases in r. □

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